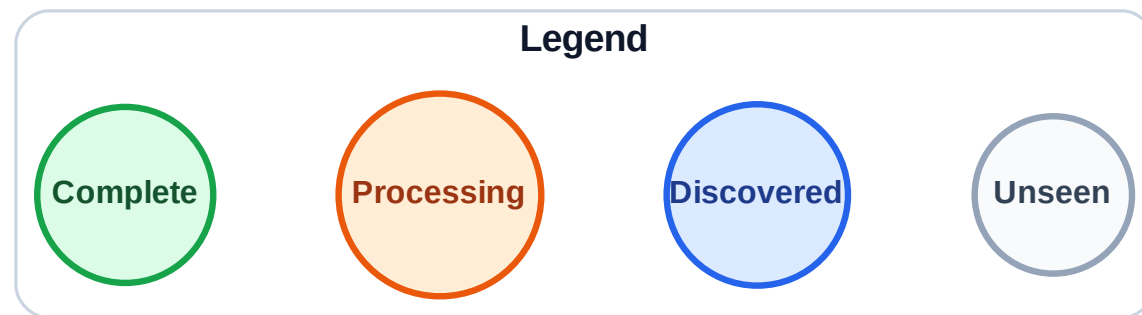
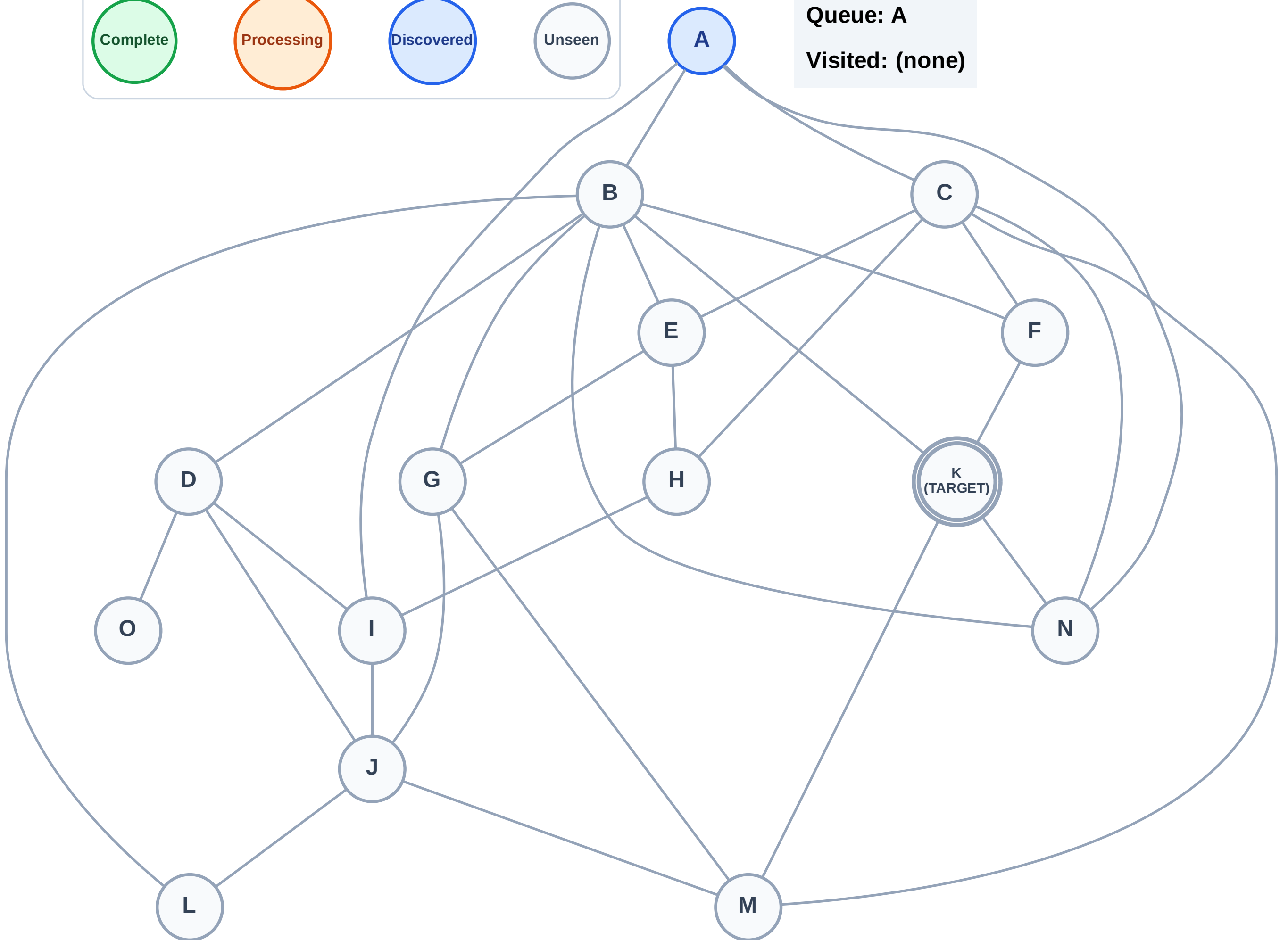


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

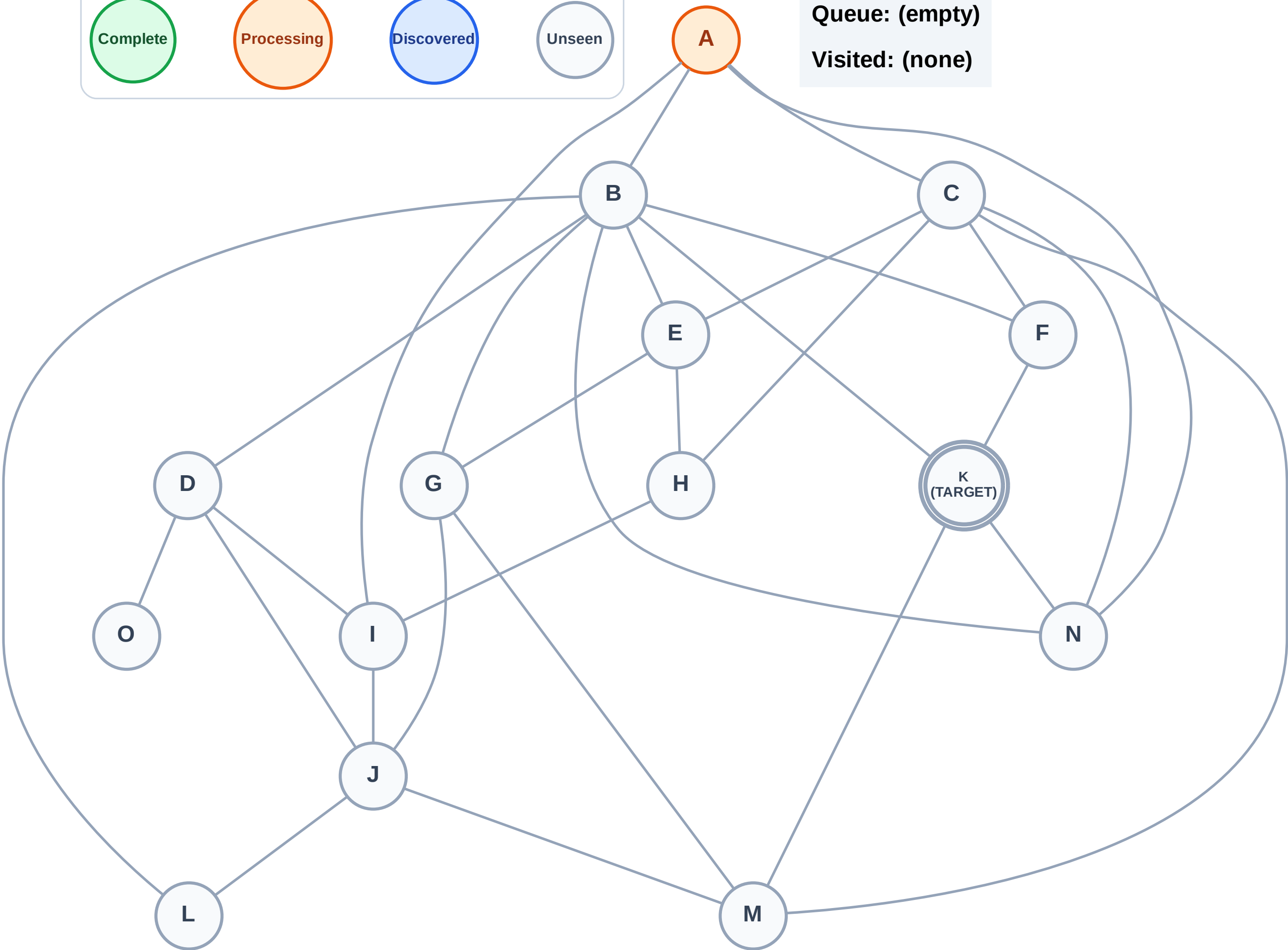
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)

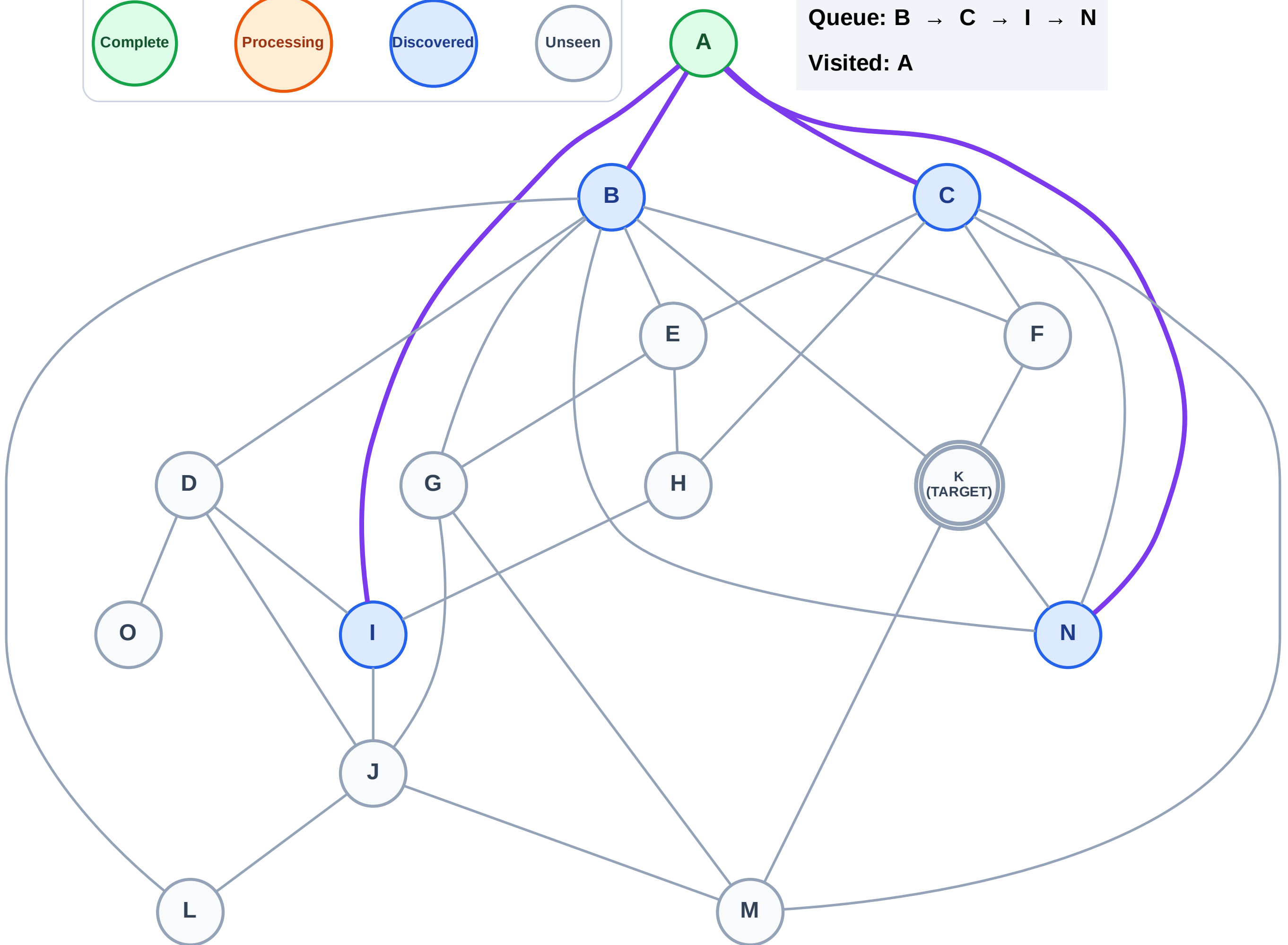


Step 3: Discover B, C, I, N from A

Legend

Unseen

Visited: A



BFS Traversal

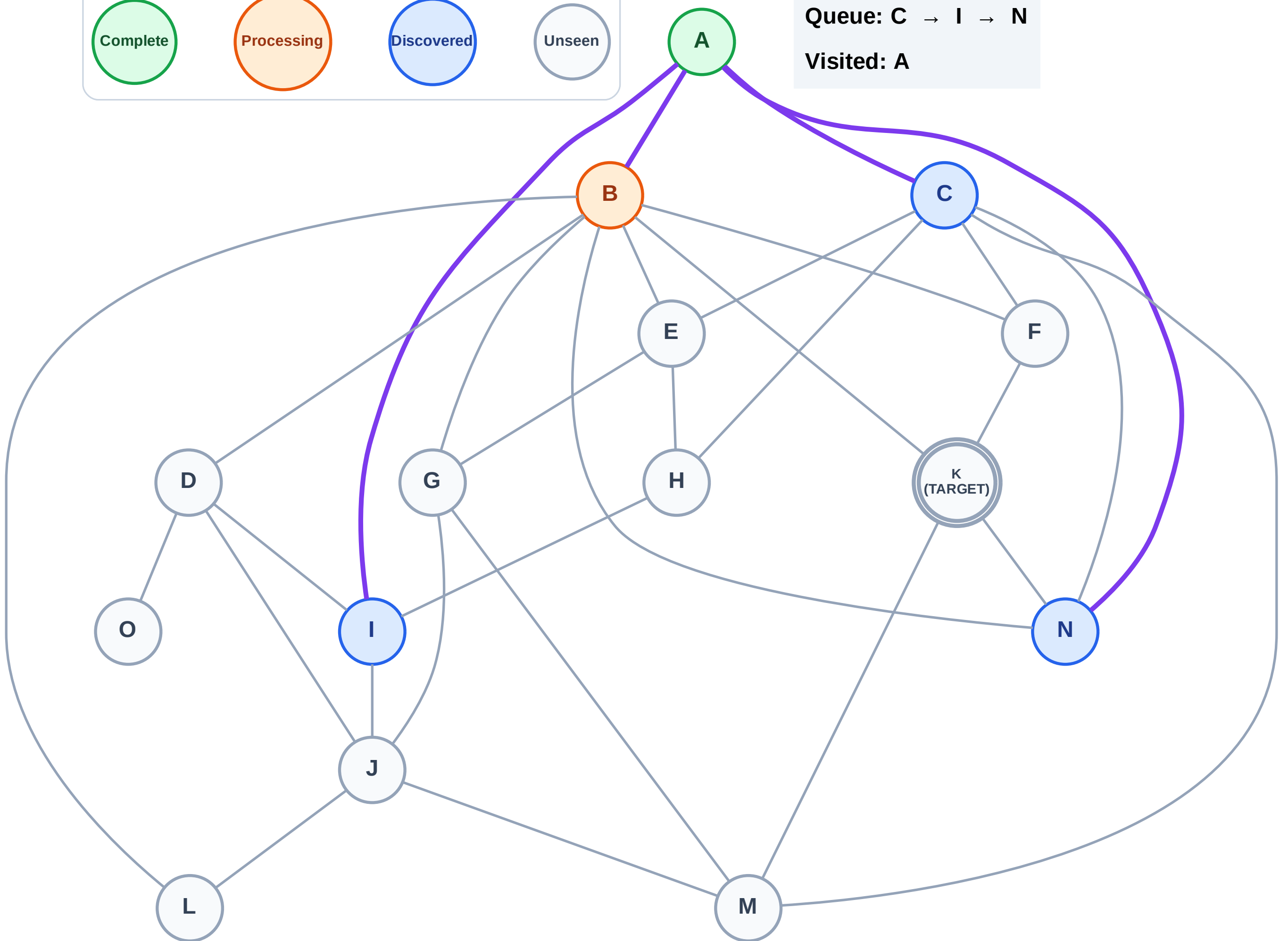
Step 4: Process node B

Legend



Queue: C → I → N

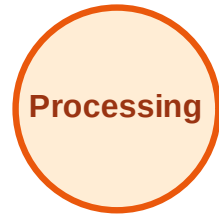
Visited: A



BFS Traversal

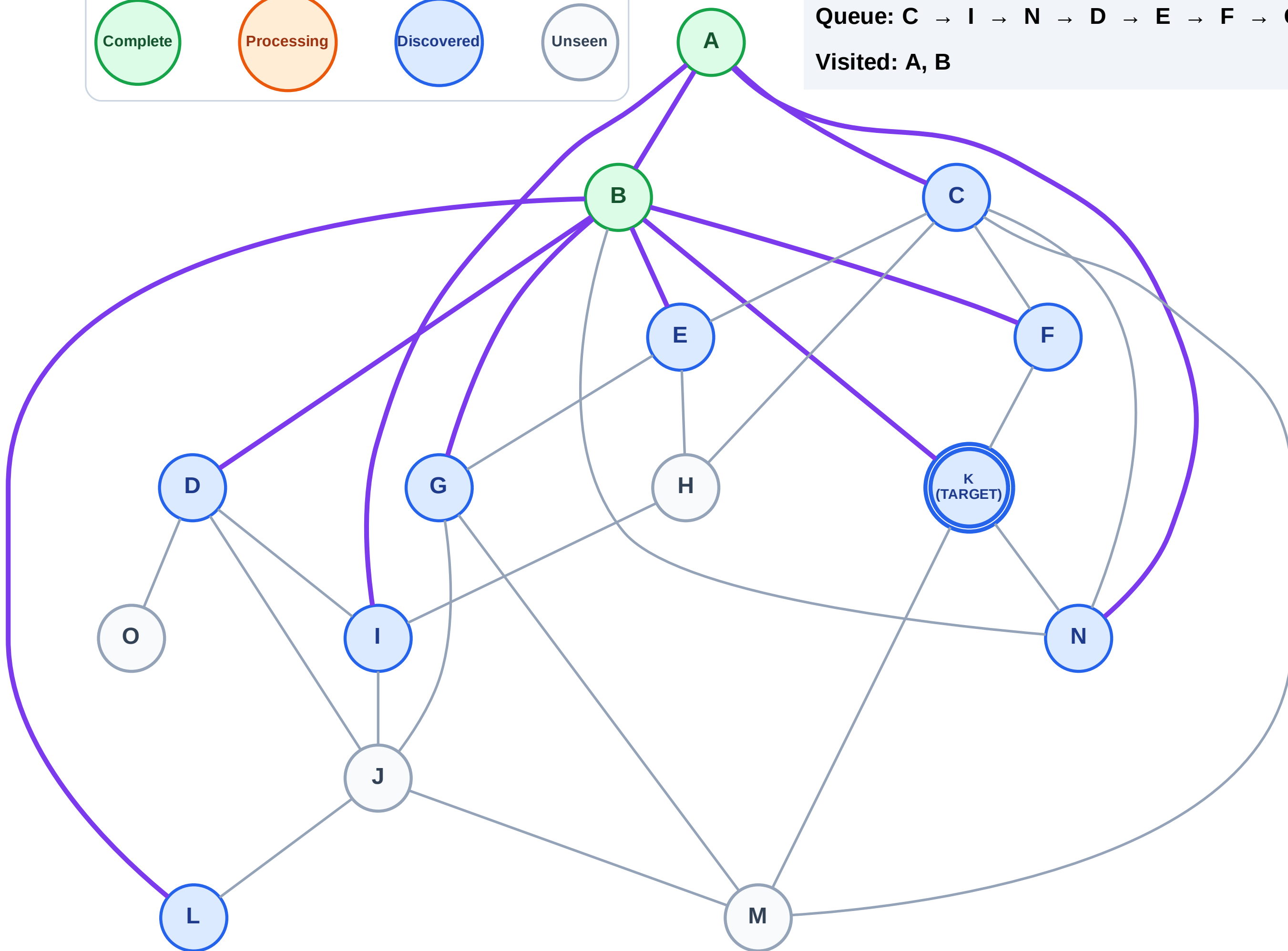
Step 5: Discover D, E, F, G, K, L from B

Legend



Queue: C → I → N → D → E → F → G → K → L

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

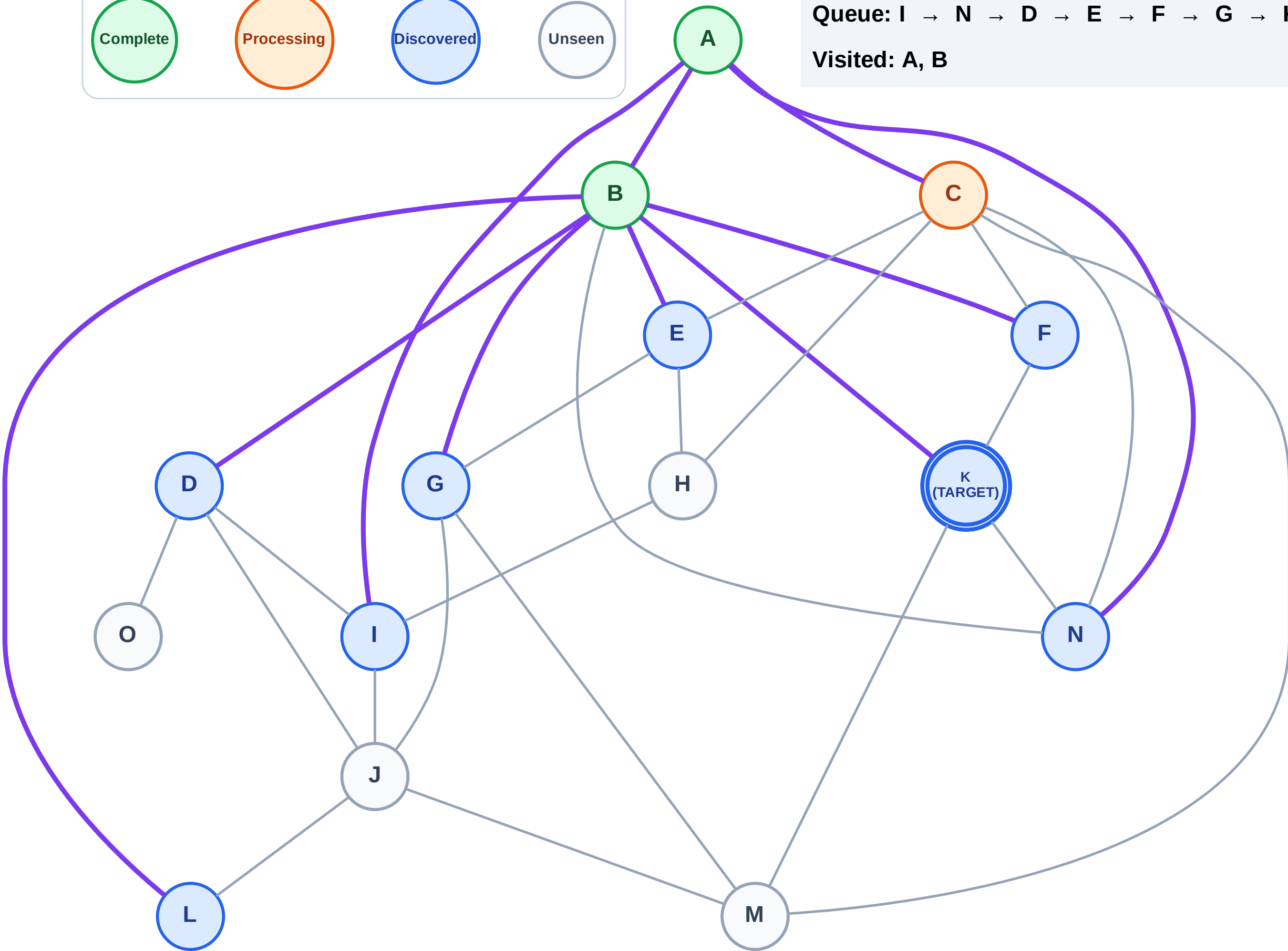
Processing

Discovered

Unseen

Queue: I → N → D → E → F → G → K → L

Visited: A, B



BFS Traversal

Step 7: Discover H, M from C

Legend

Complete

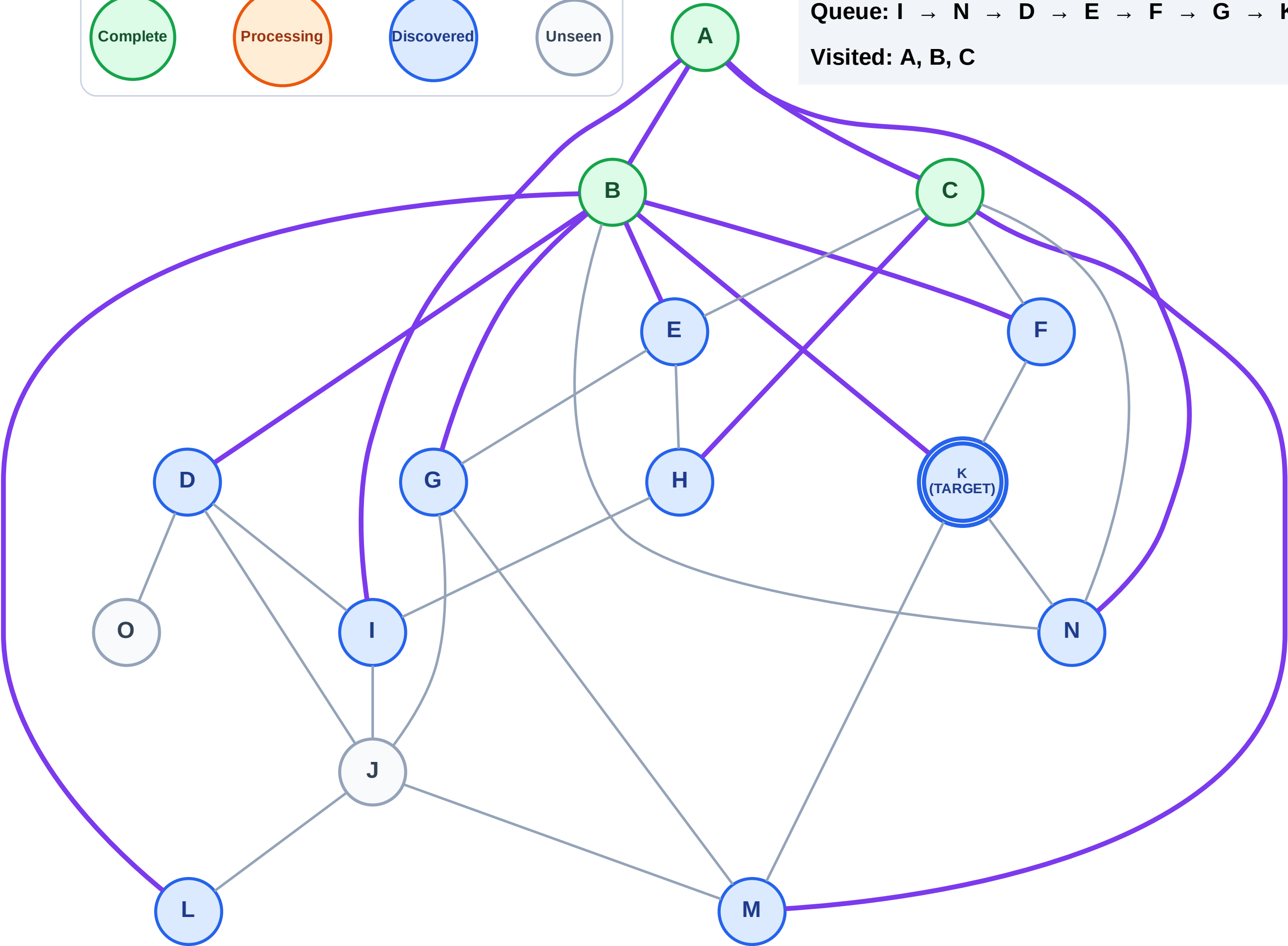
Processing

Discovered

Unseen

Queue: I → N → D → E → F → G → K → L → H → M

Visited: A, B, C



BFS Traversal

Step 8: Process node I

Legend

Complete

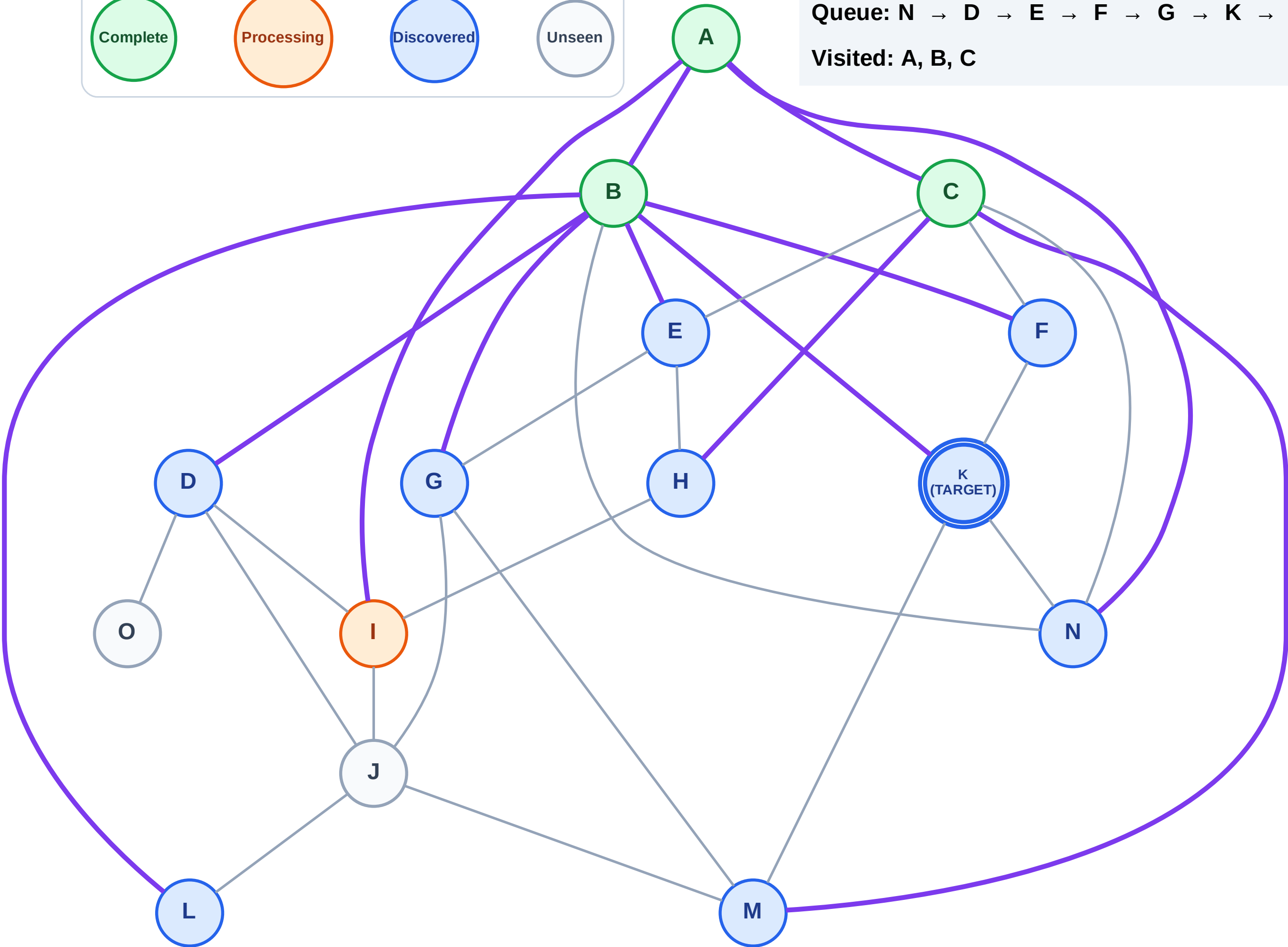
Processing

Discovered

Unseen

Queue: N → D → E → F → G → K → L → H → M

Visited: A, B, C



BFS Traversal

Step 9: Discover J from I

Legend

Complete

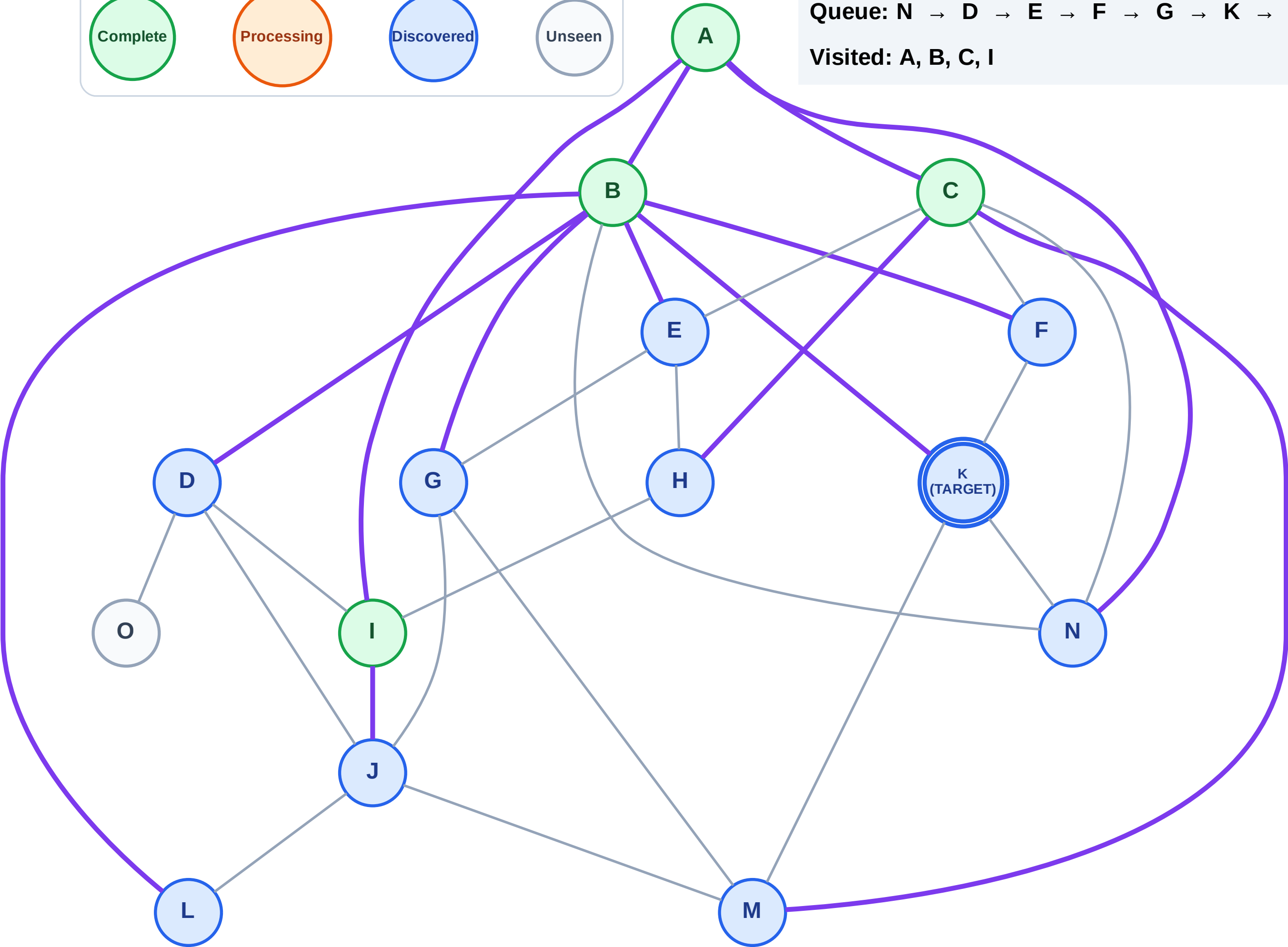
Processing

Discovered

Unseen

Queue: N → D → E → F → G → K → L → H → M → J

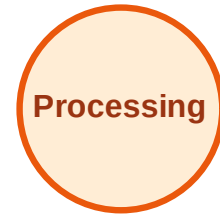
Visited: A, B, C, I



BFS Traversal

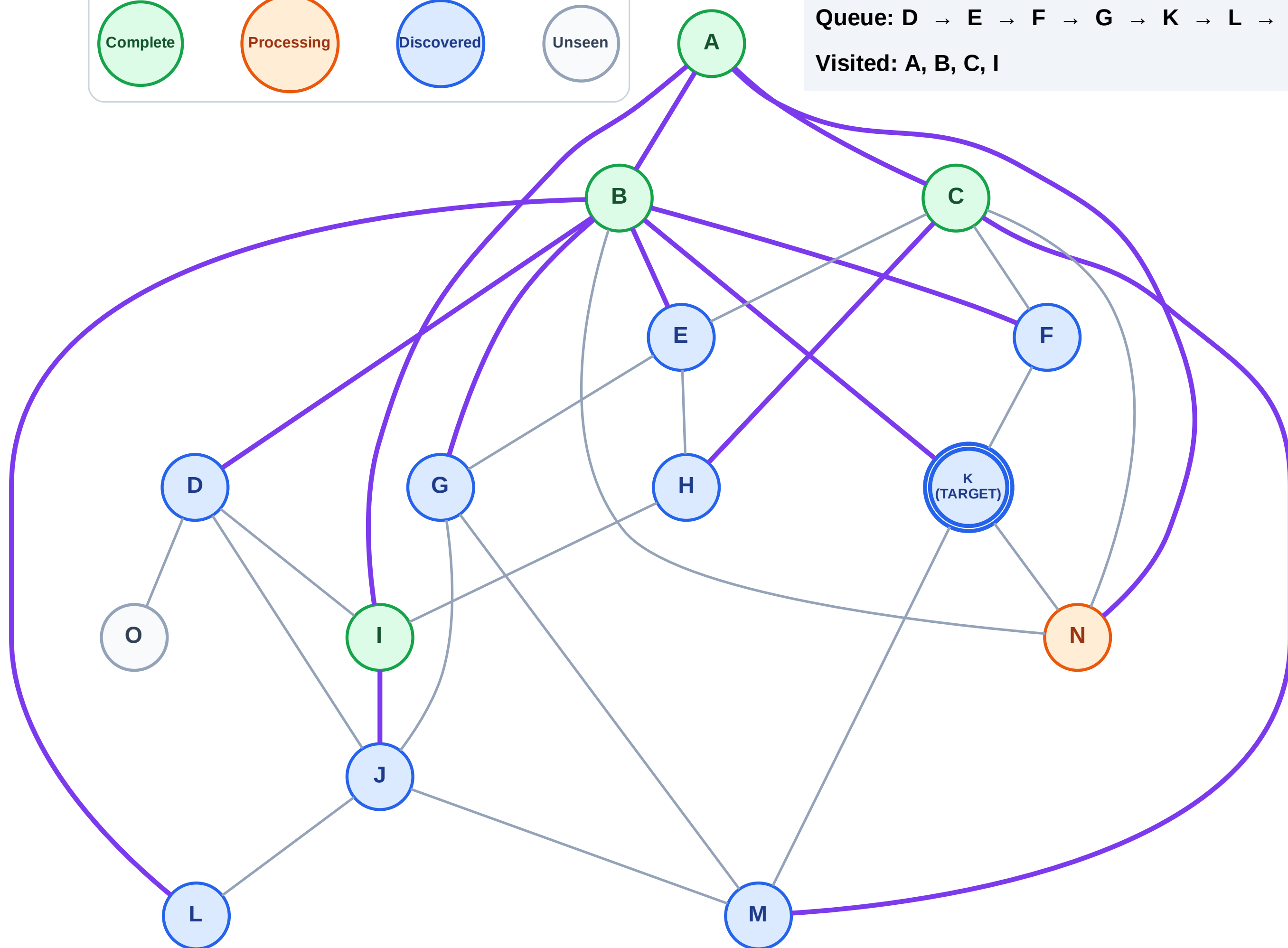
Step 10: Process node N

Legend



Queue: D → E → F → G → K → L → H → M → J

Visited: A, B, C, I



BFS Traversal

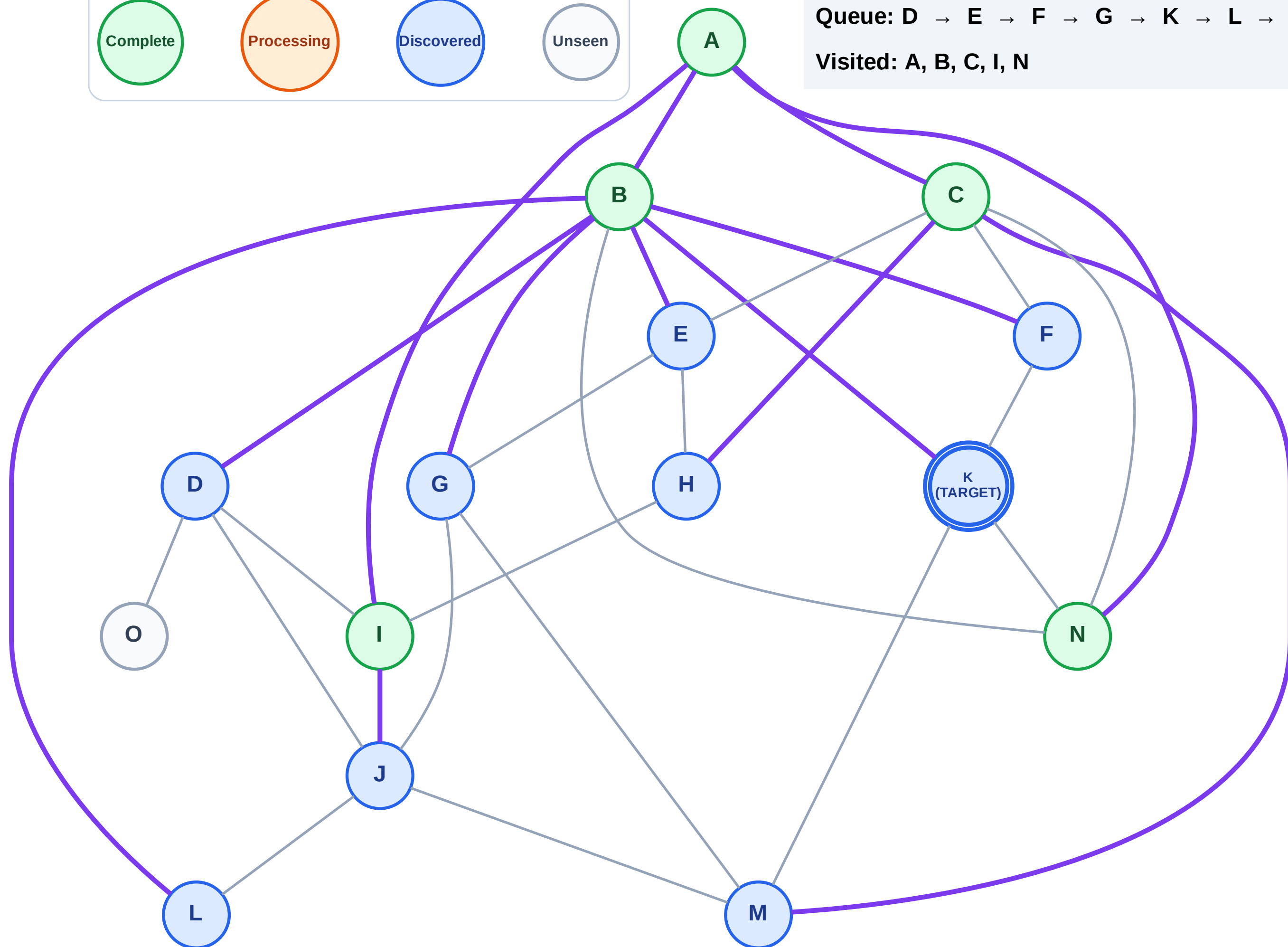
Step 11: No new neighbors from N

Legend



Queue: D → E → F → G → K → L → H → M → J

Visited: A, B, C, I, N



BFS Traversal

Step 12: Process node D

Legend

Complete

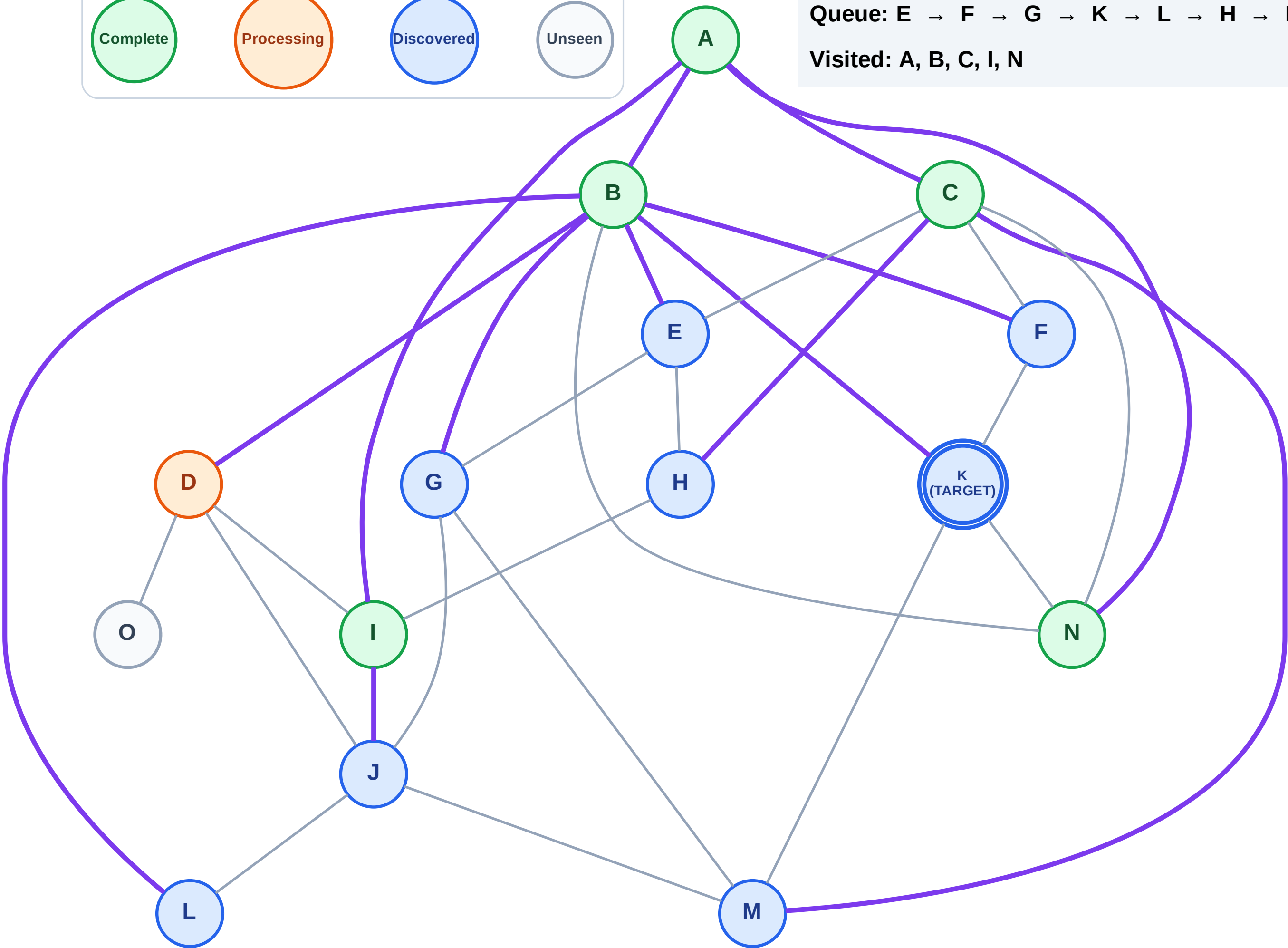
Processing

Discovered

Unseen

Queue: E → F → G → K → L → H → M → J

Visited: A, B, C, I, N



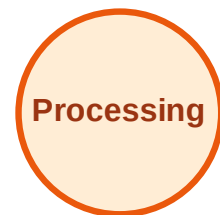
BFS Traversal

Step 13: Discover O from D

Legend



Complete



Processing



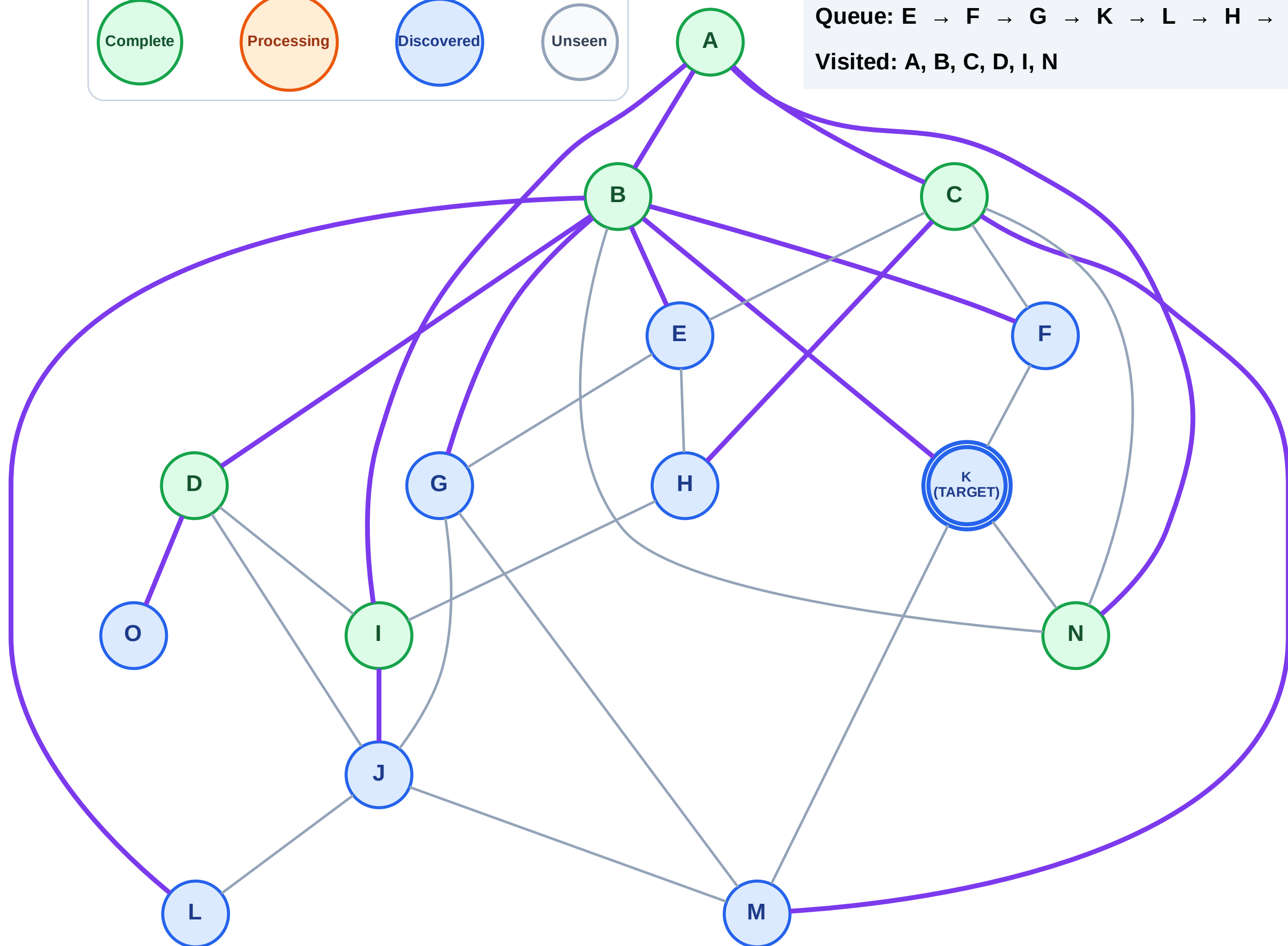
Discovered



Unseen

Queue: E → F → G → K → L → H → M → J → O

Visited: A, B, C, D, I, N



BFS Traversal

Step 14: Process node E

Legend

Complete

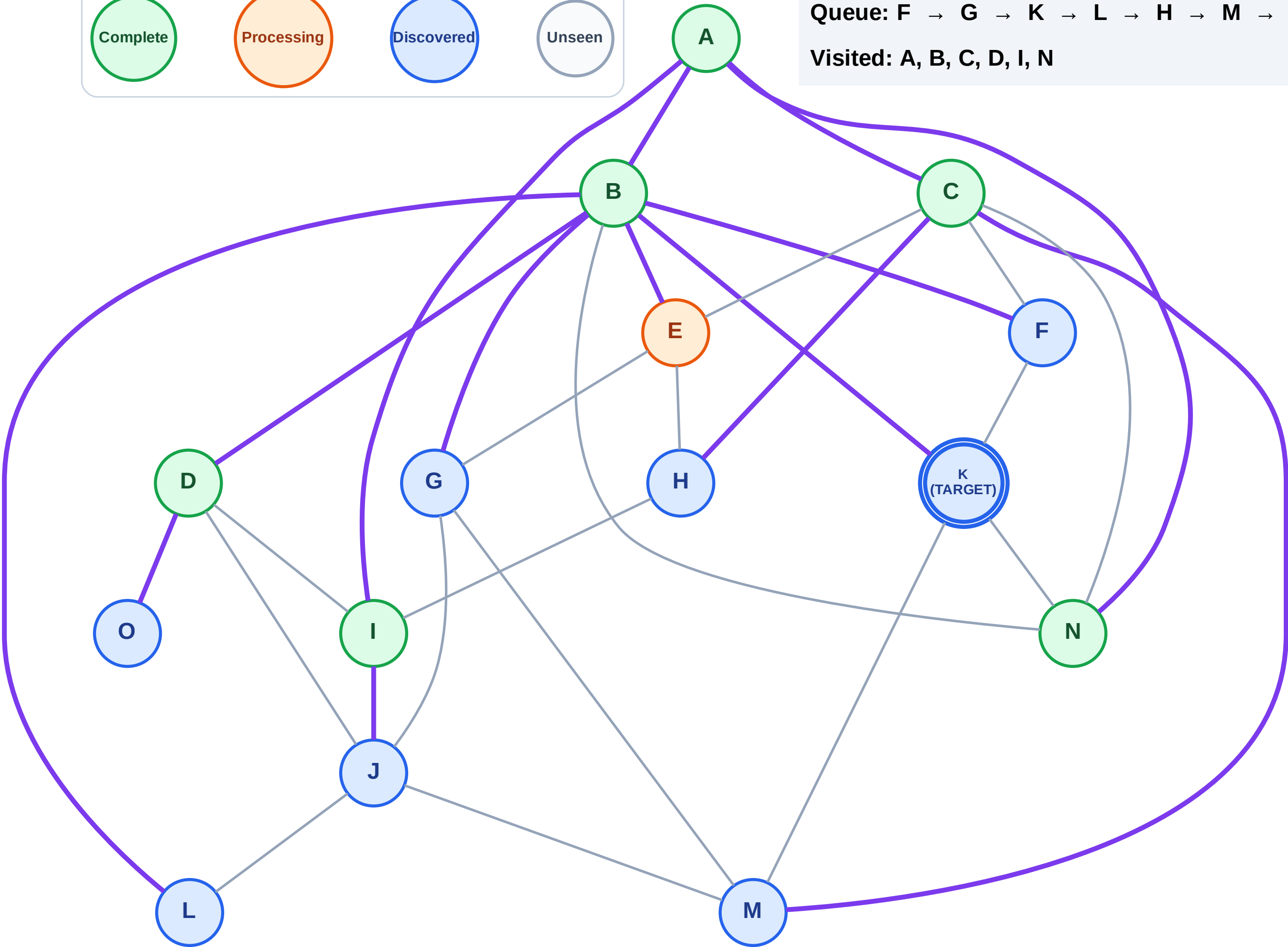
Processing

Discovered

Unseen

Queue: F → G → K → L → H → M → J → O

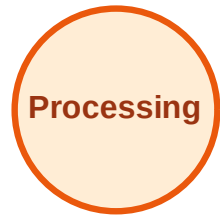
Visited: A, B, C, D, I, N



BFS Traversal

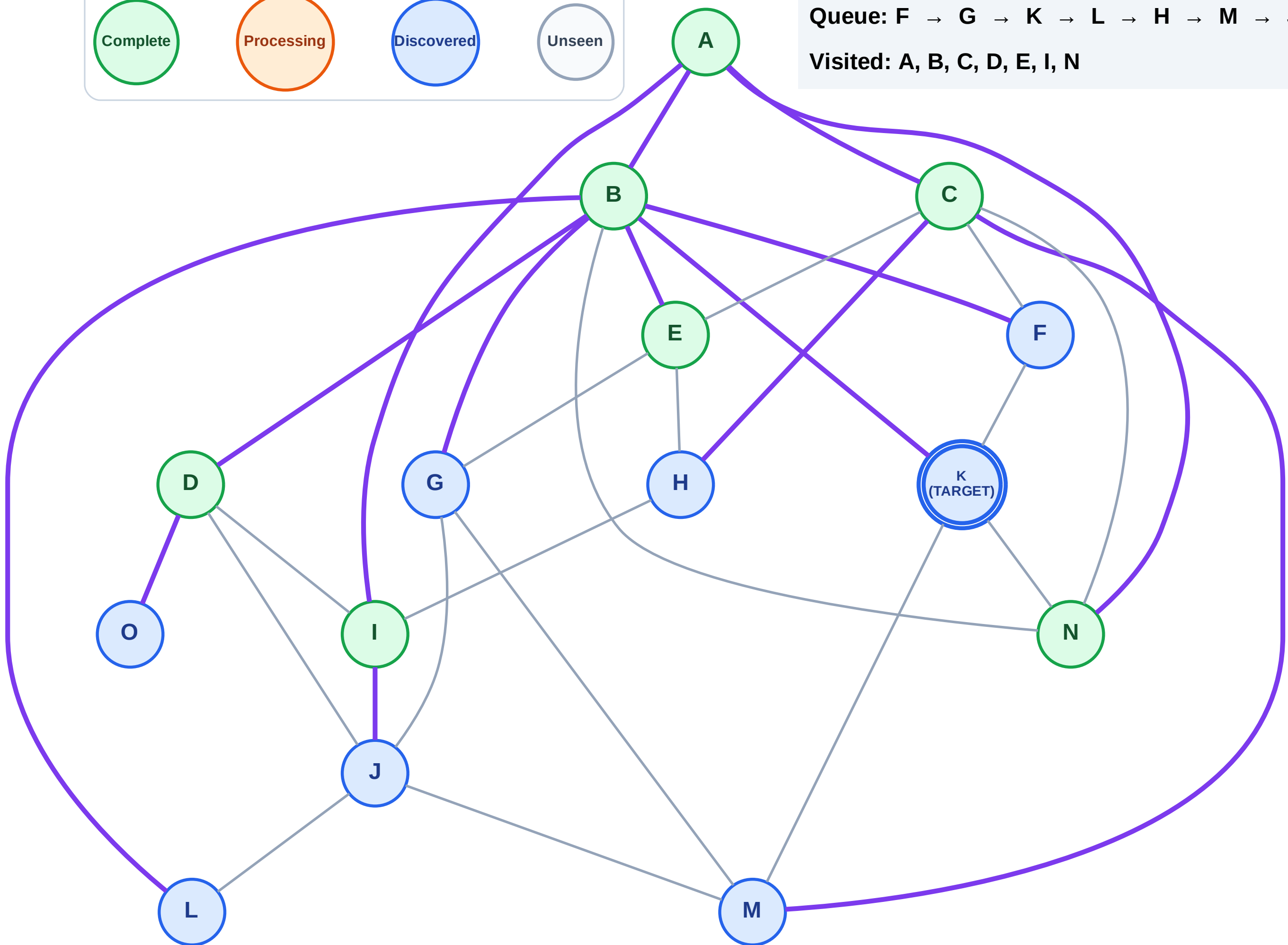
Step 15: No new neighbors from E

Legend



Queue: F → G → K → L → H → M → J → O

Visited: A, B, C, D, E, I, N



BFS Traversal

Step 16: Process node F

Legend

Complete

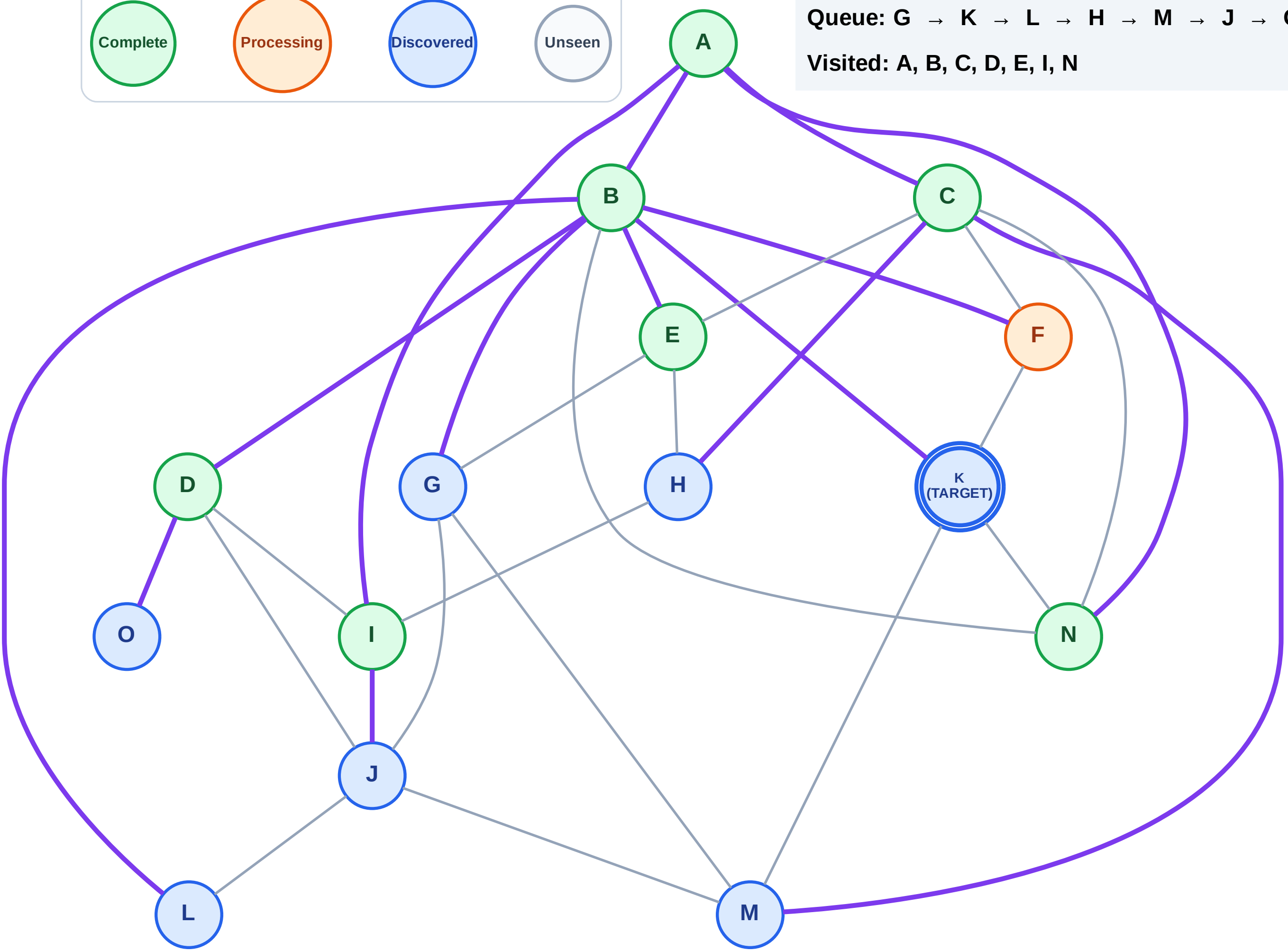
Processing

Discovered

Unseen

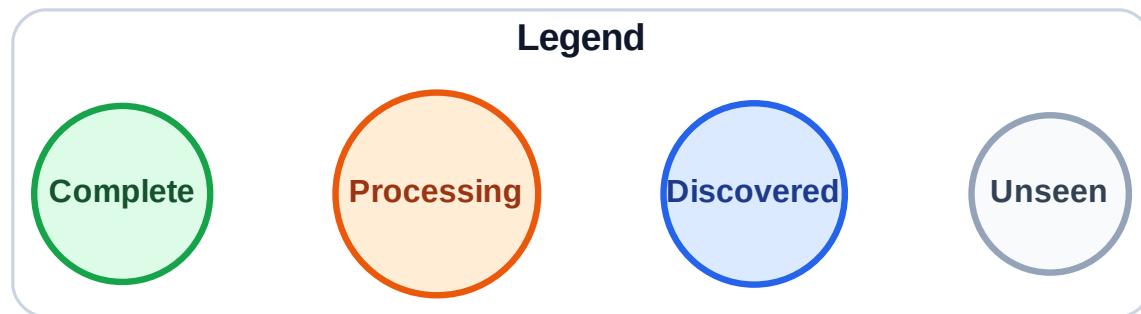
Queue: G → K → L → H → M → J → O

Visited: A, B, C, D, E, I, N



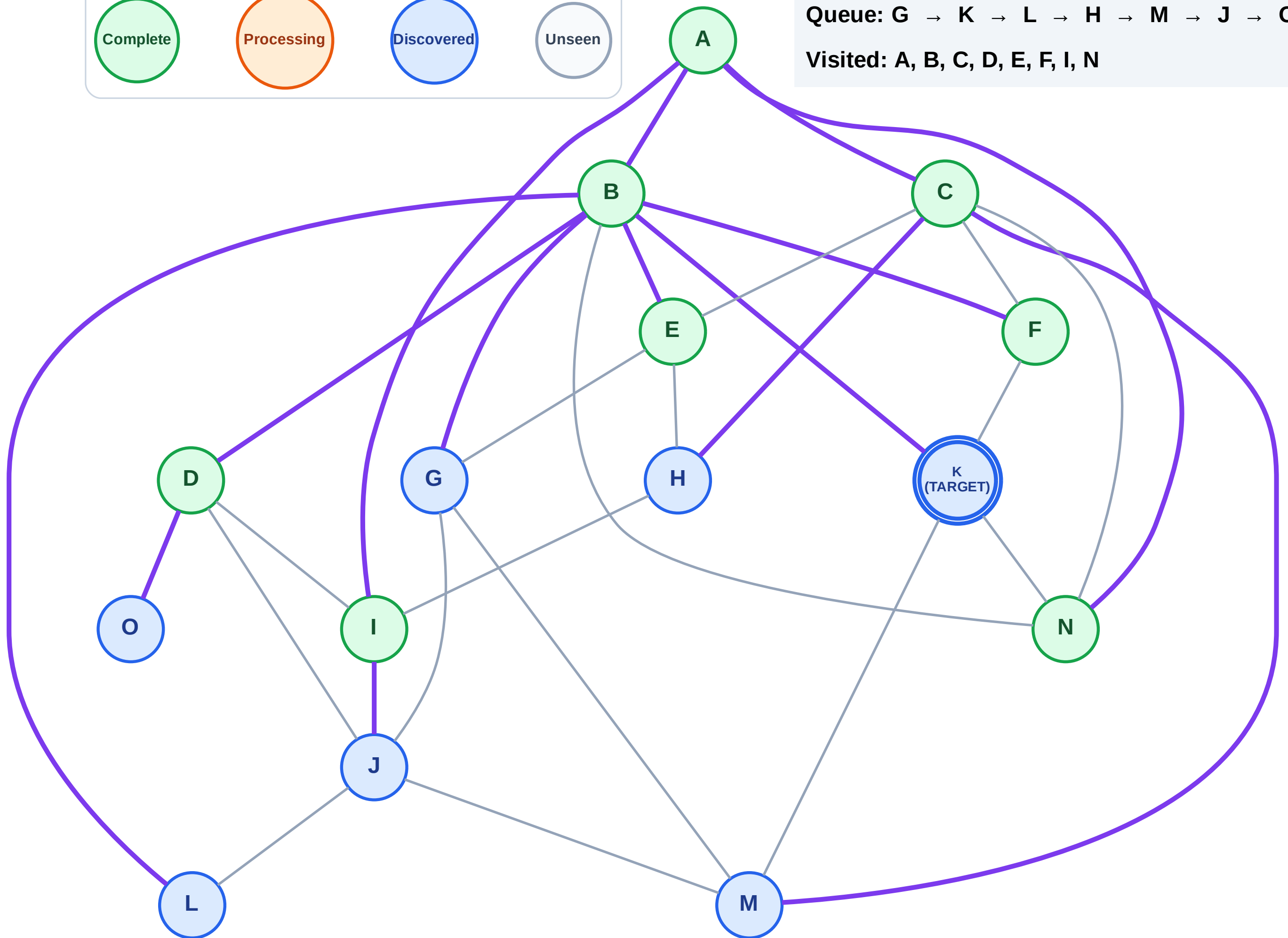
BFS Traversal

Step 17: No new neighbors from F



Queue: G → K → L → H → M → J → O

Visited: A, B, C, D, E, F, I, N



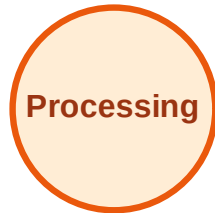
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



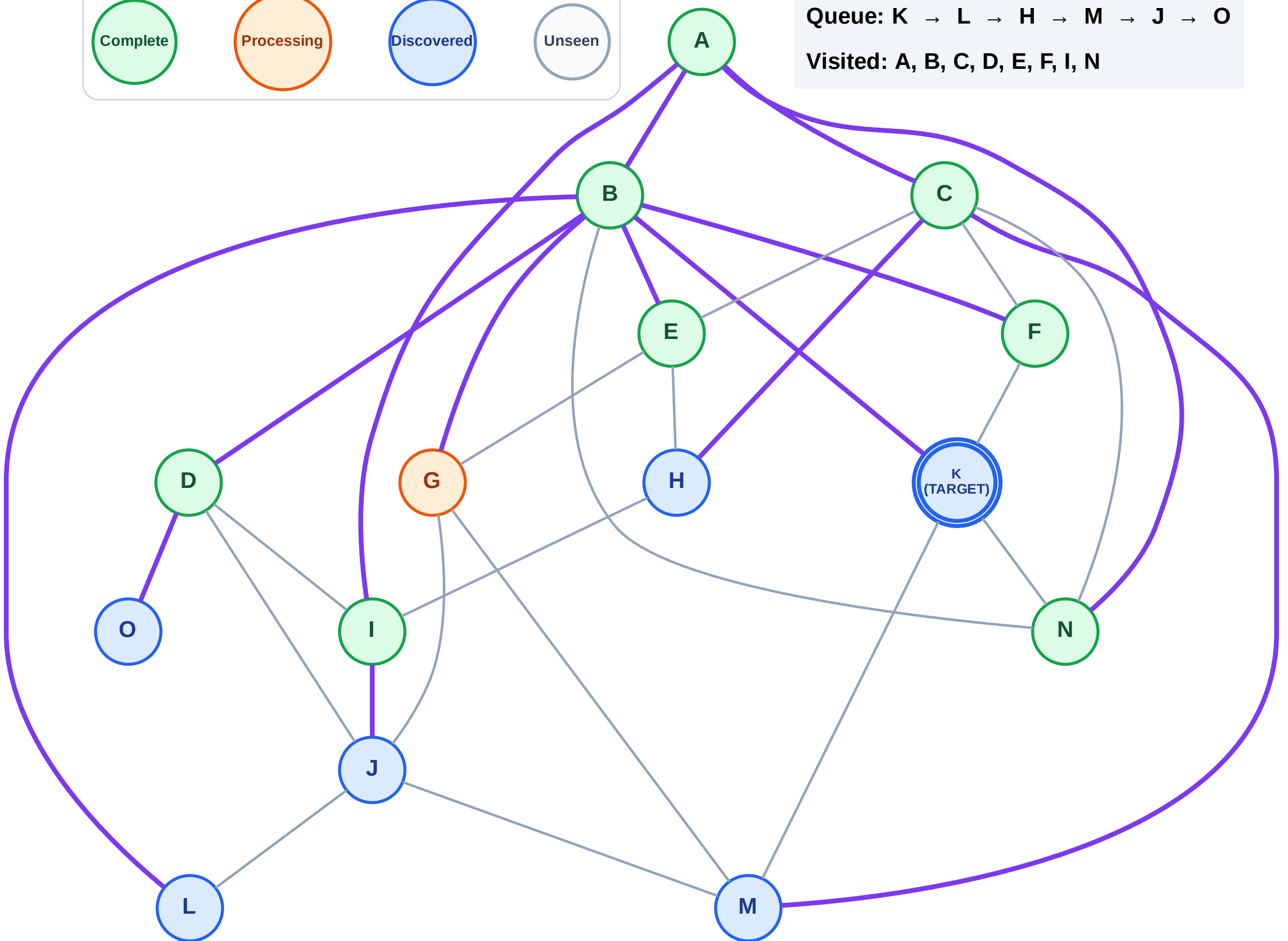
Discovered



Unseen

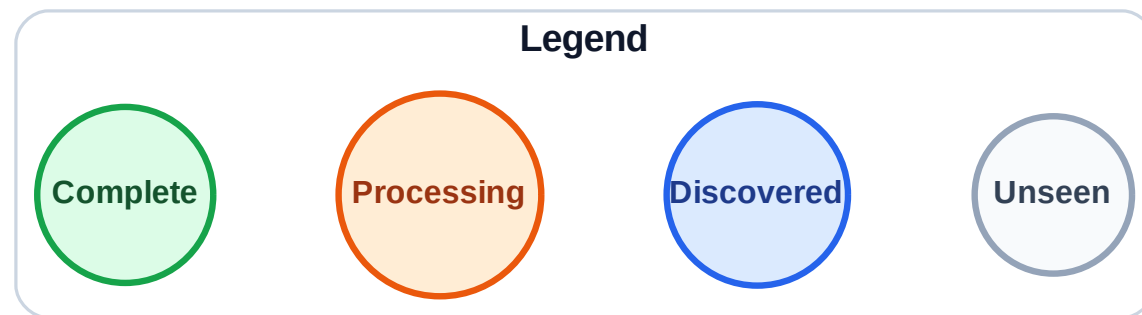
Queue: K → L → H → M → J → O

Visited: A, B, C, D, E, F, I, N



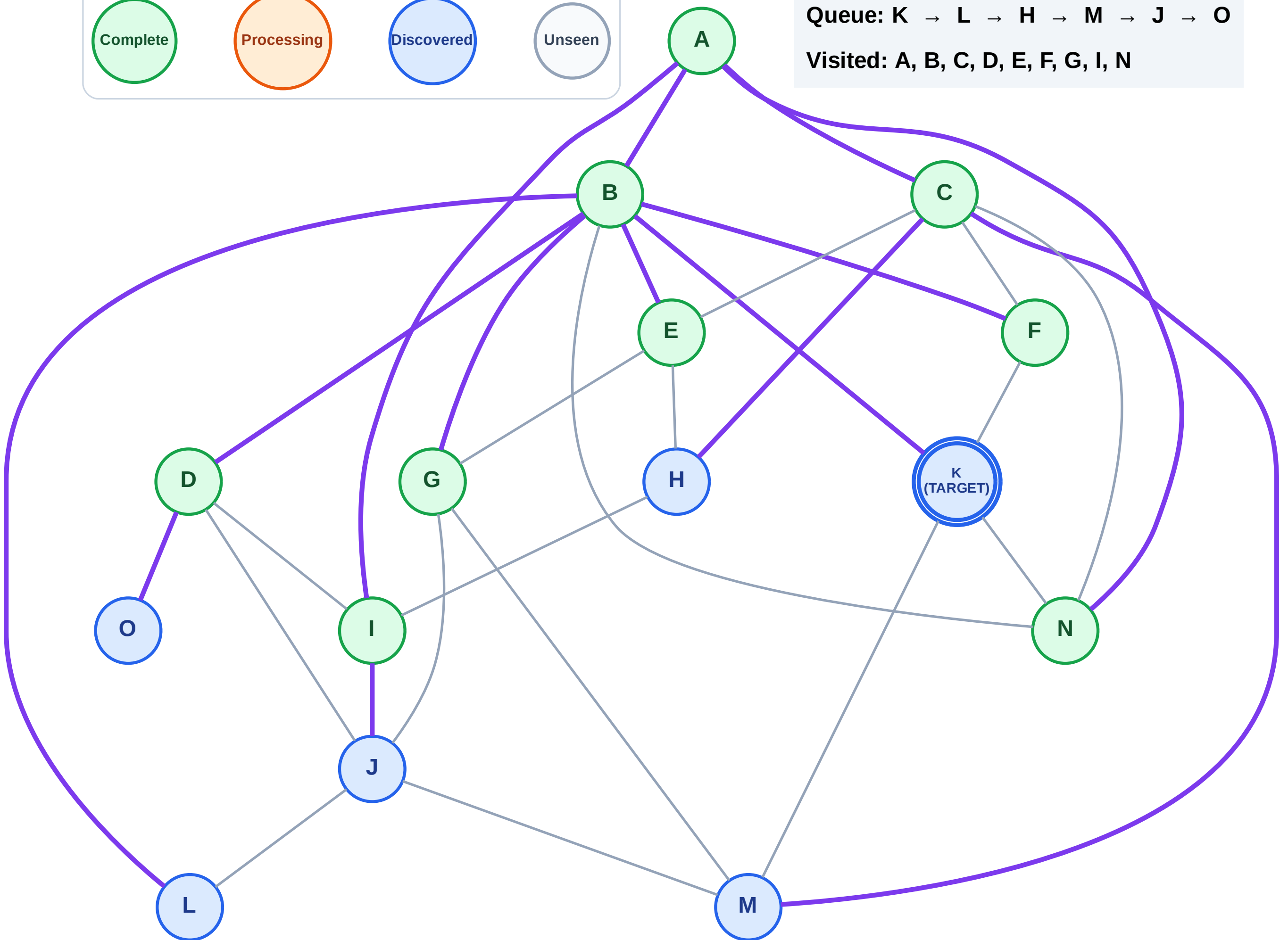
BFS Traversal

Step 19: No new neighbors from G



Queue: K → L → H → M → J → O

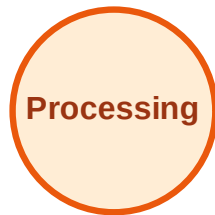
Visited: A, B, C, D, E, F, G, I, N



BFS Traversal

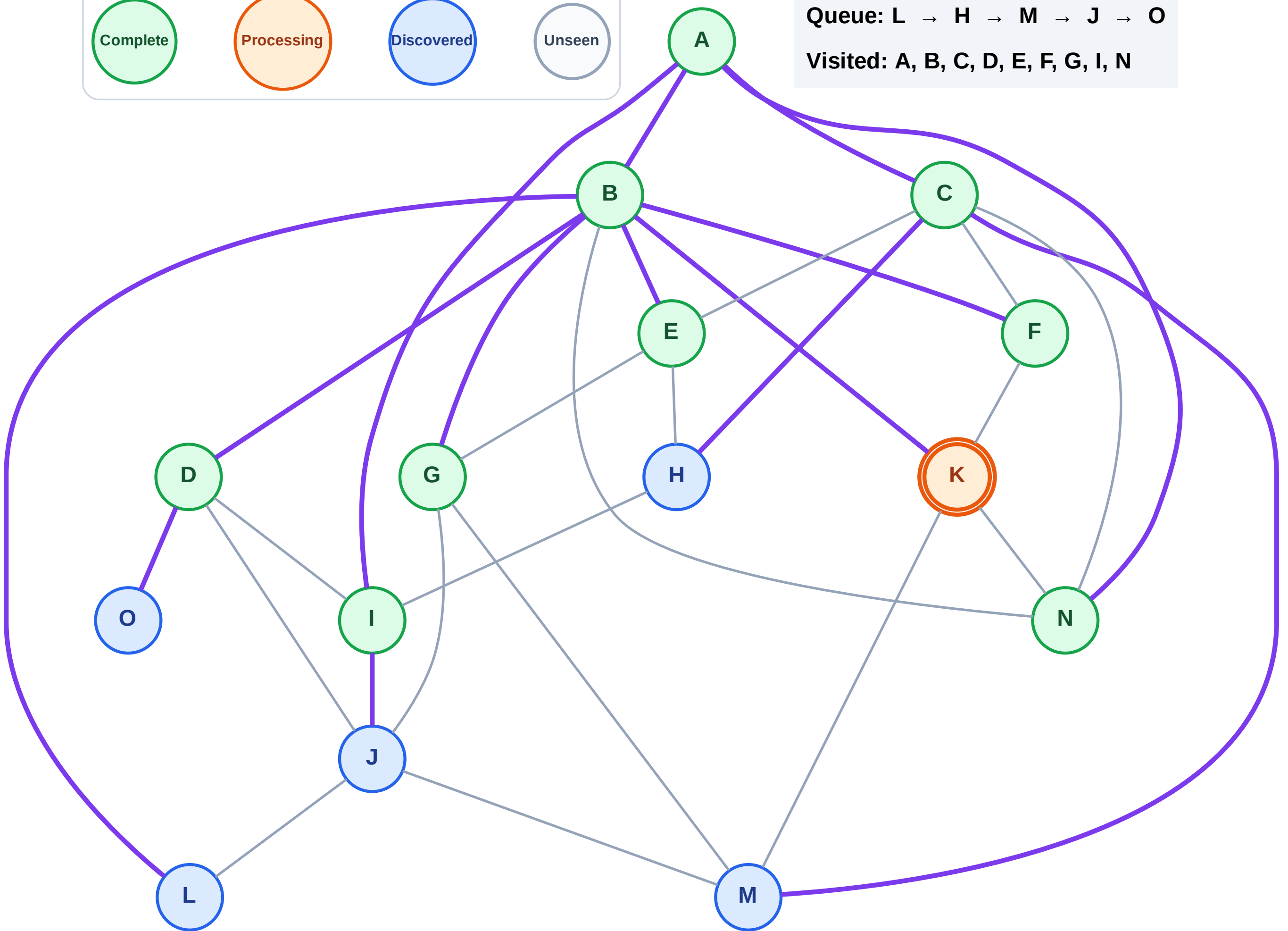
Step 20: Process node K

Legend



Queue: L → H → M → J → O

Visited: A, B, C, D, E, F, G, I, N



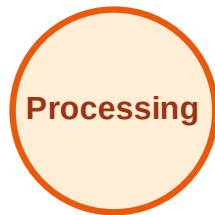
BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



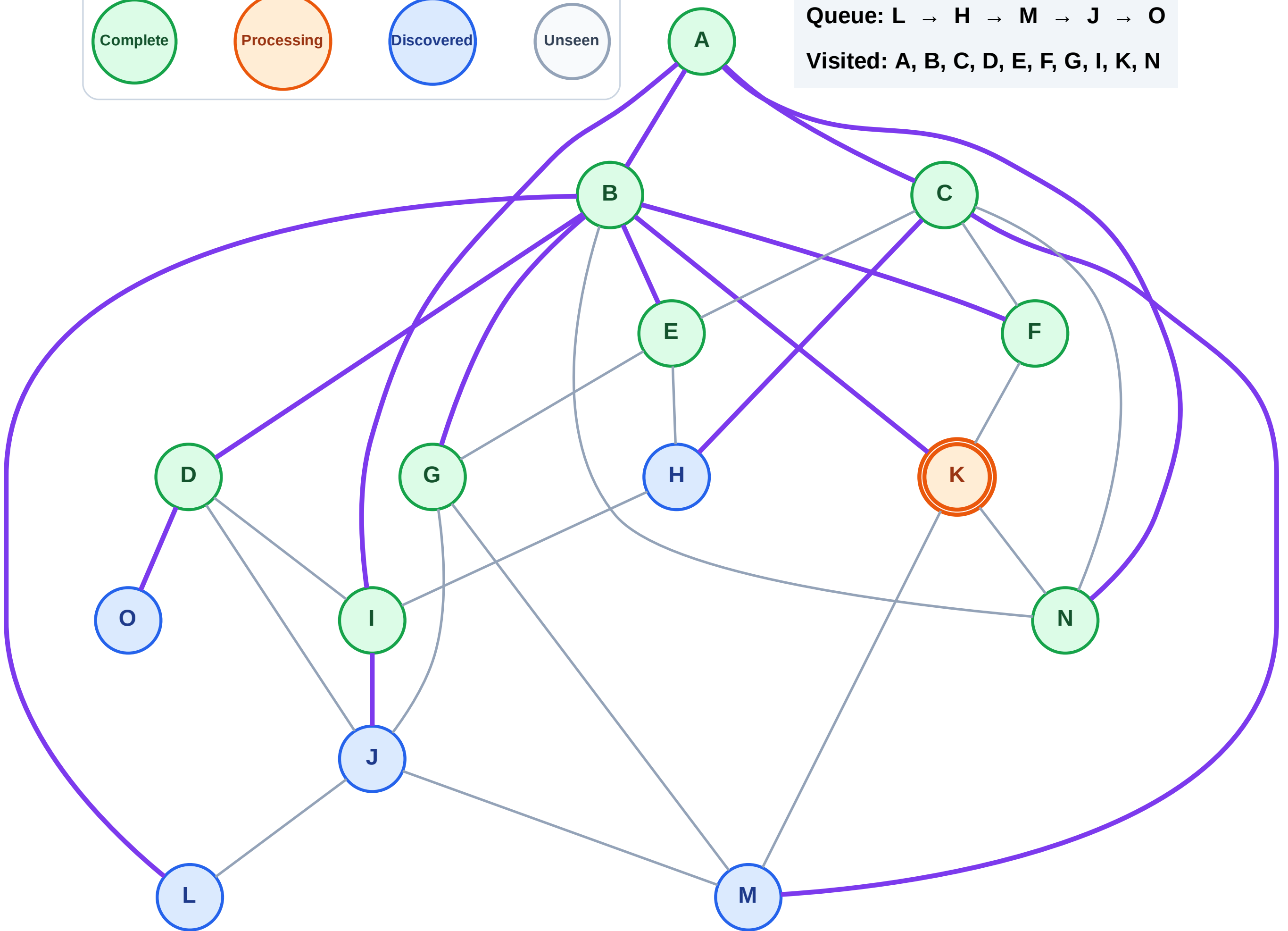
Discovered



Unseen

Queue: L → H → M → J → O

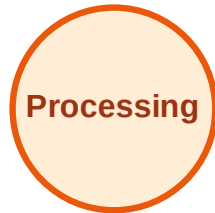
Visited: A, B, C, D, E, F, G, I, K, N



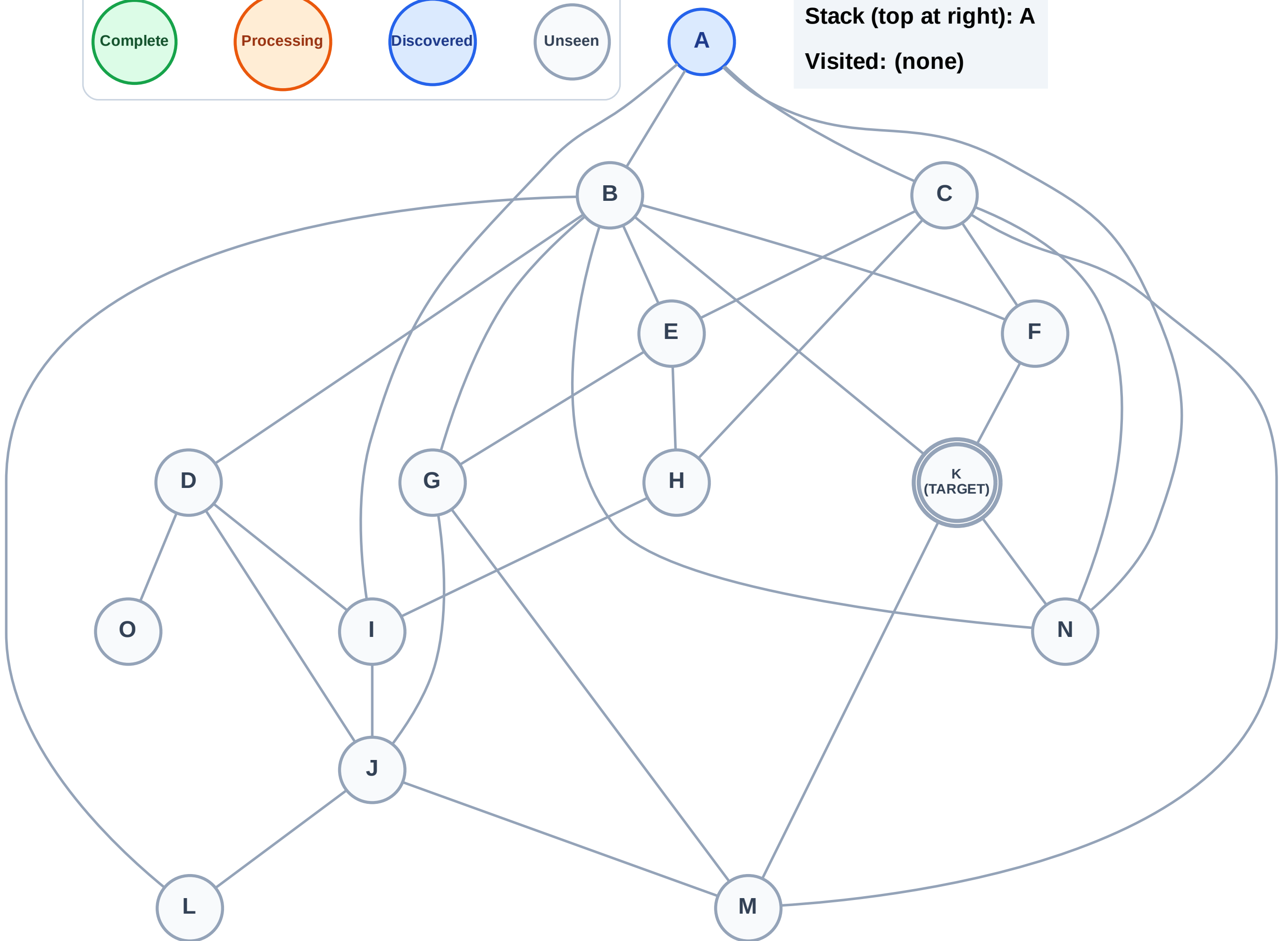
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

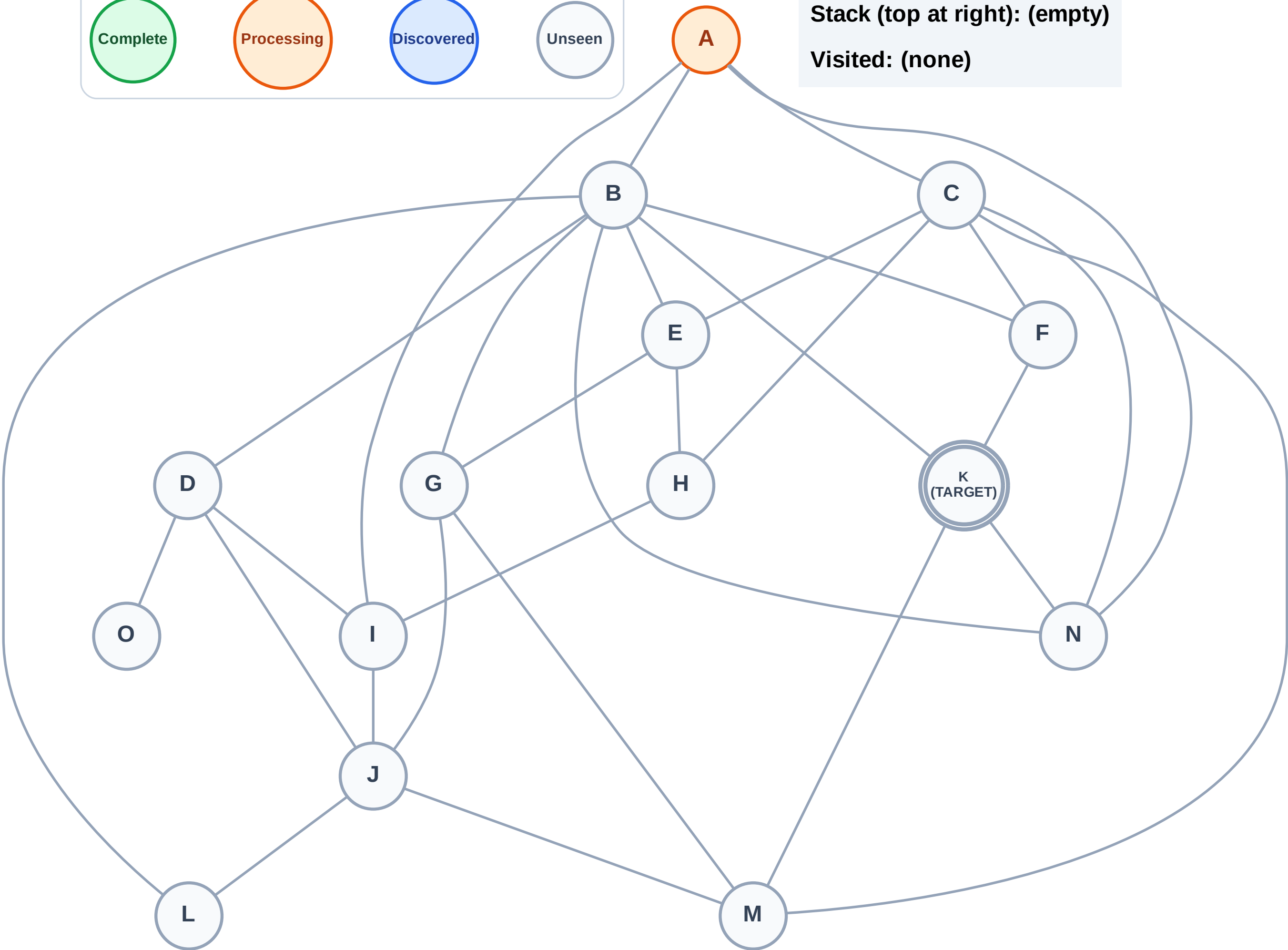
Complete

Processing

Discovered

Unseen

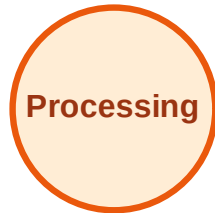
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

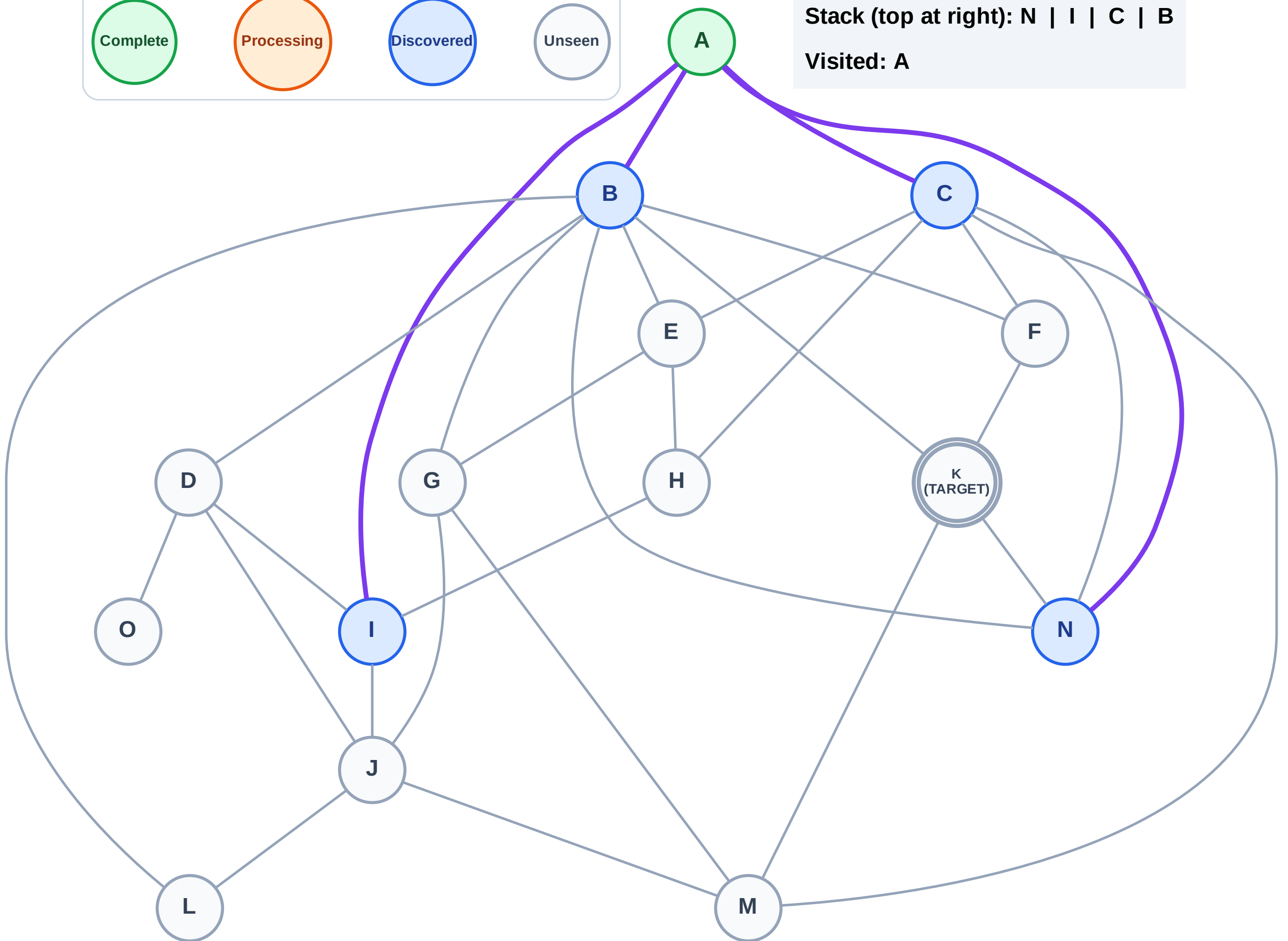
Step 3: Discover N, I, C, B from A

Legend



Stack (top at right): N | I | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

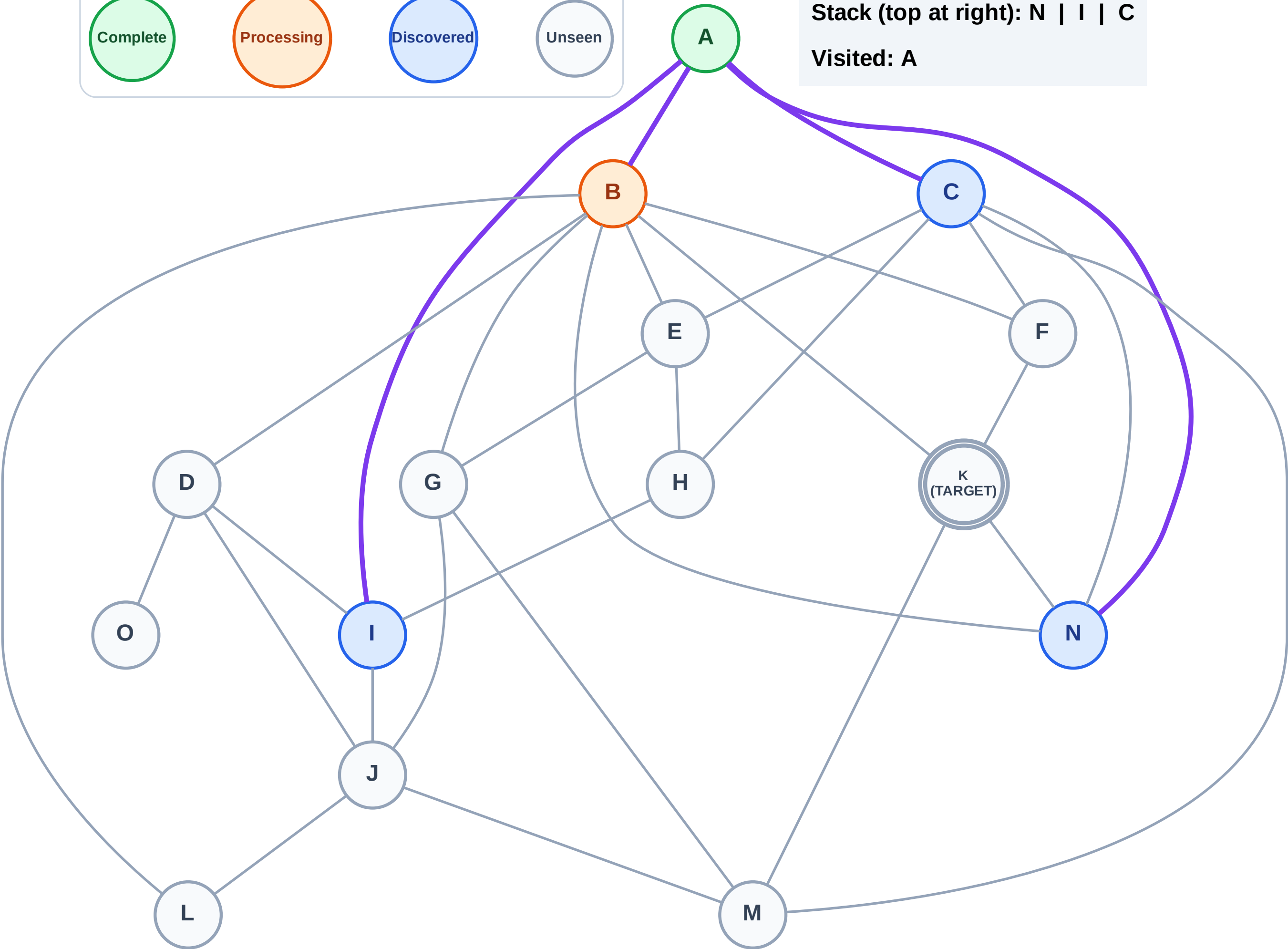
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C
Visited: A

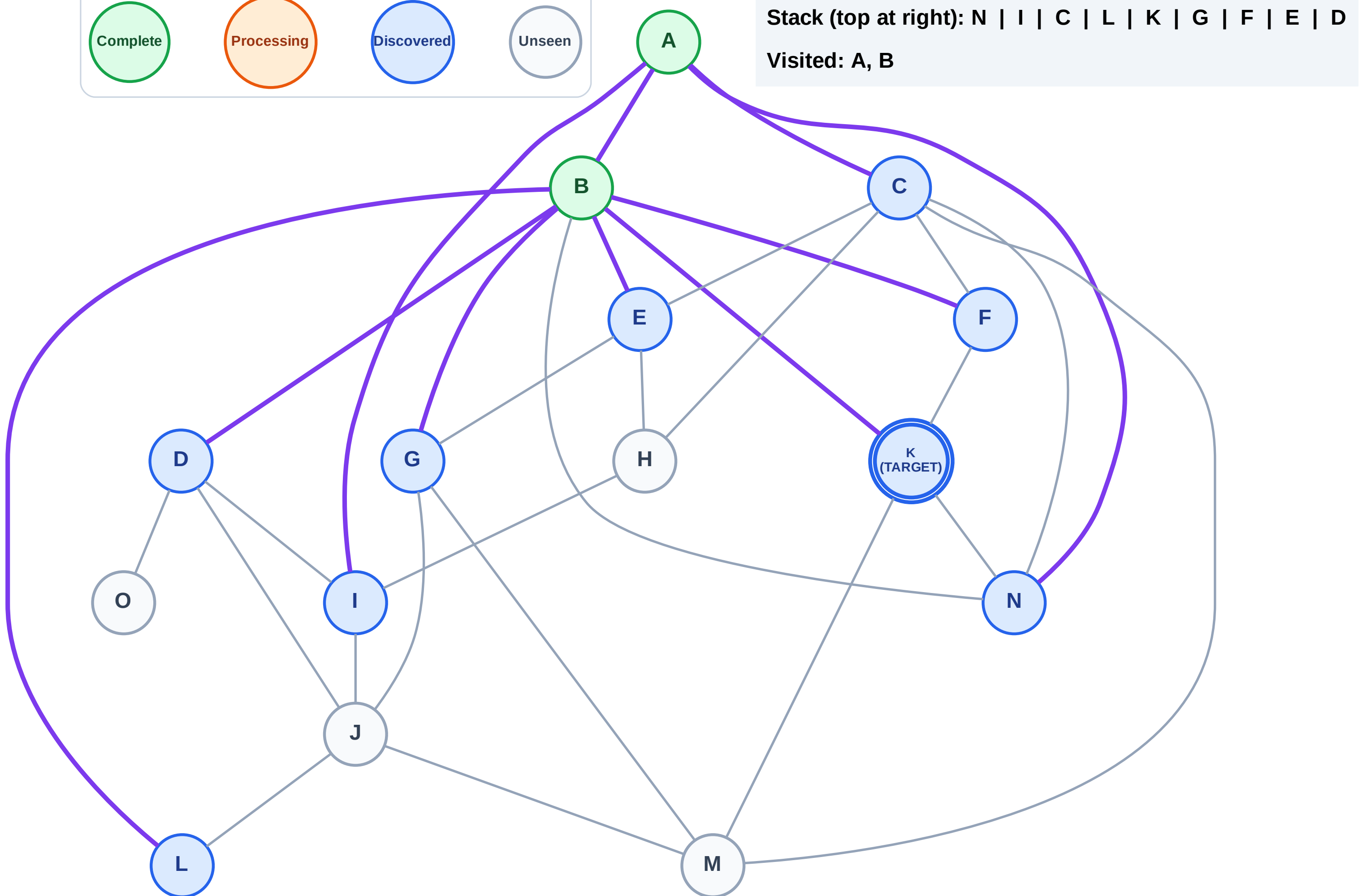


Step 5: Discover L, K, G, F, E, D from B

Legend

Unseen

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

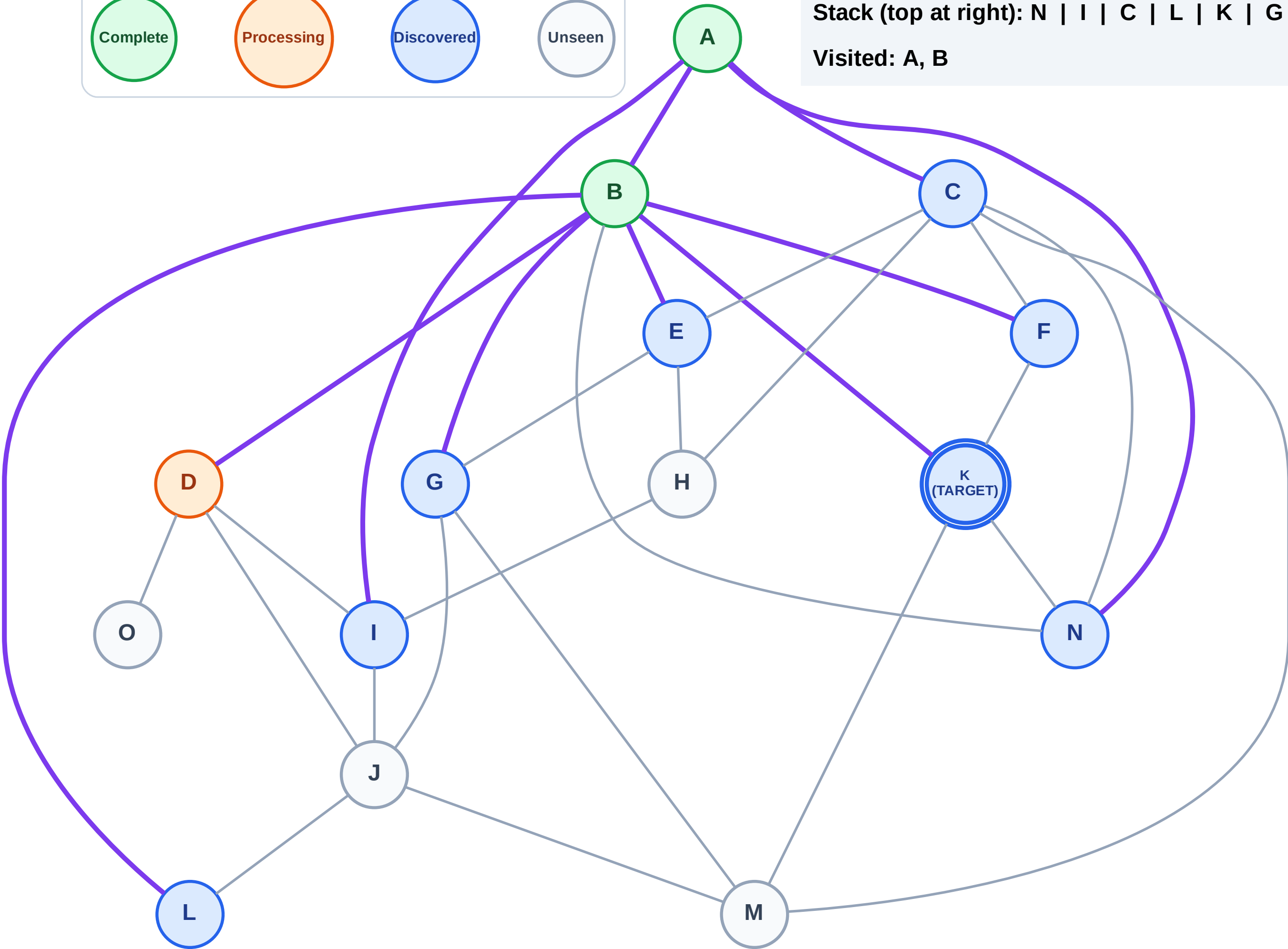
Processing

Discovered

Unseen

Stack (top at right): N | I | C | L | K | G | F | E

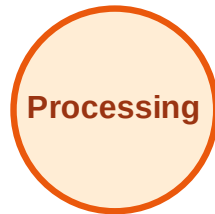
Visited: A, B



DFS Traversal

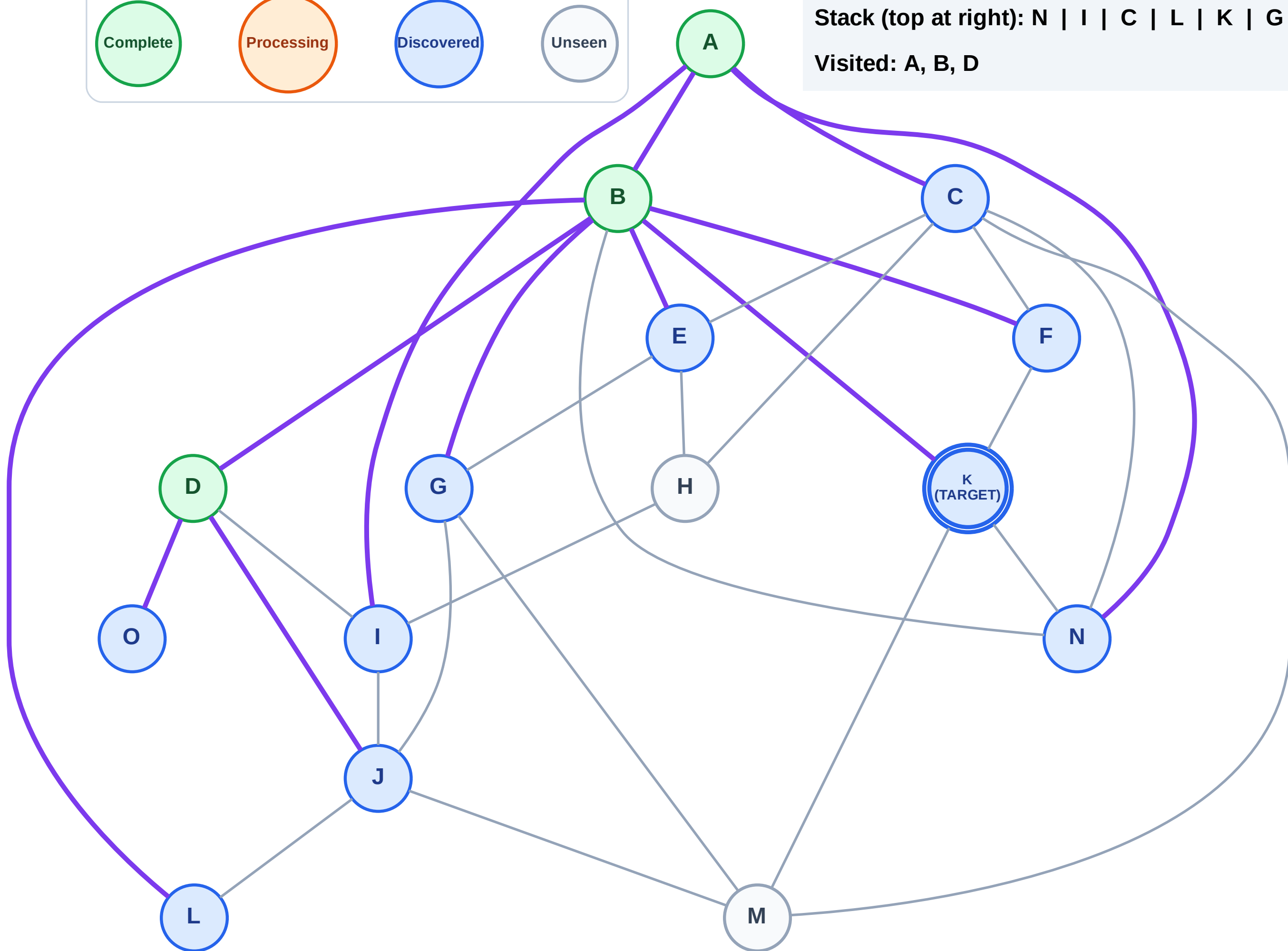
Step 7: Discover O, J from D

Legend



Stack (top at right): N | I | C | L | K | G | F | E | O | J

Visited: A, B, D



DFS Traversal

Step 8: Process node J

Legend

Complete

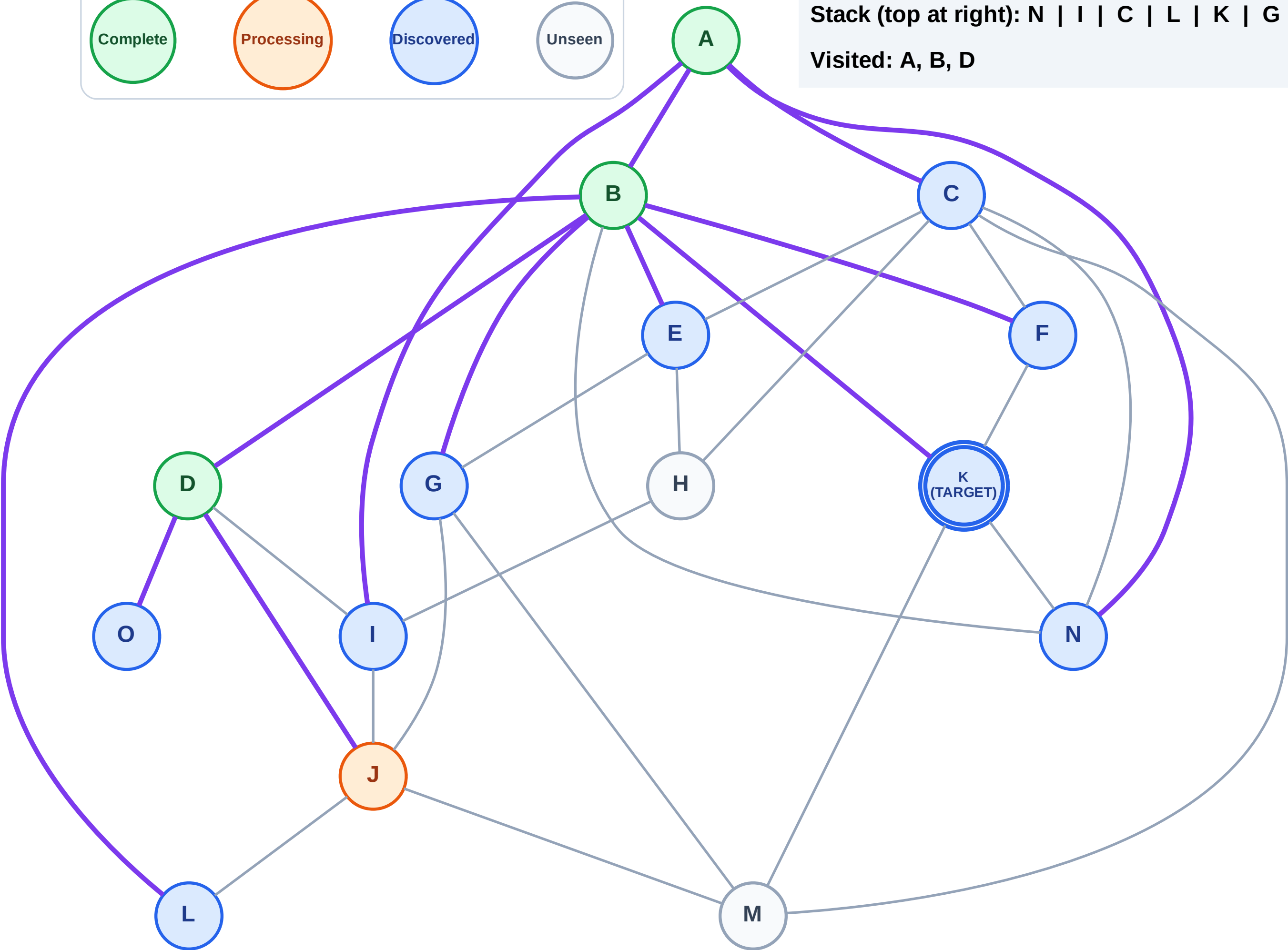
Processing

Discovered

Unseen

Stack (top at right): N | I | C | L | K | G | F | E | O

Visited: A, B, D



DFS Traversal

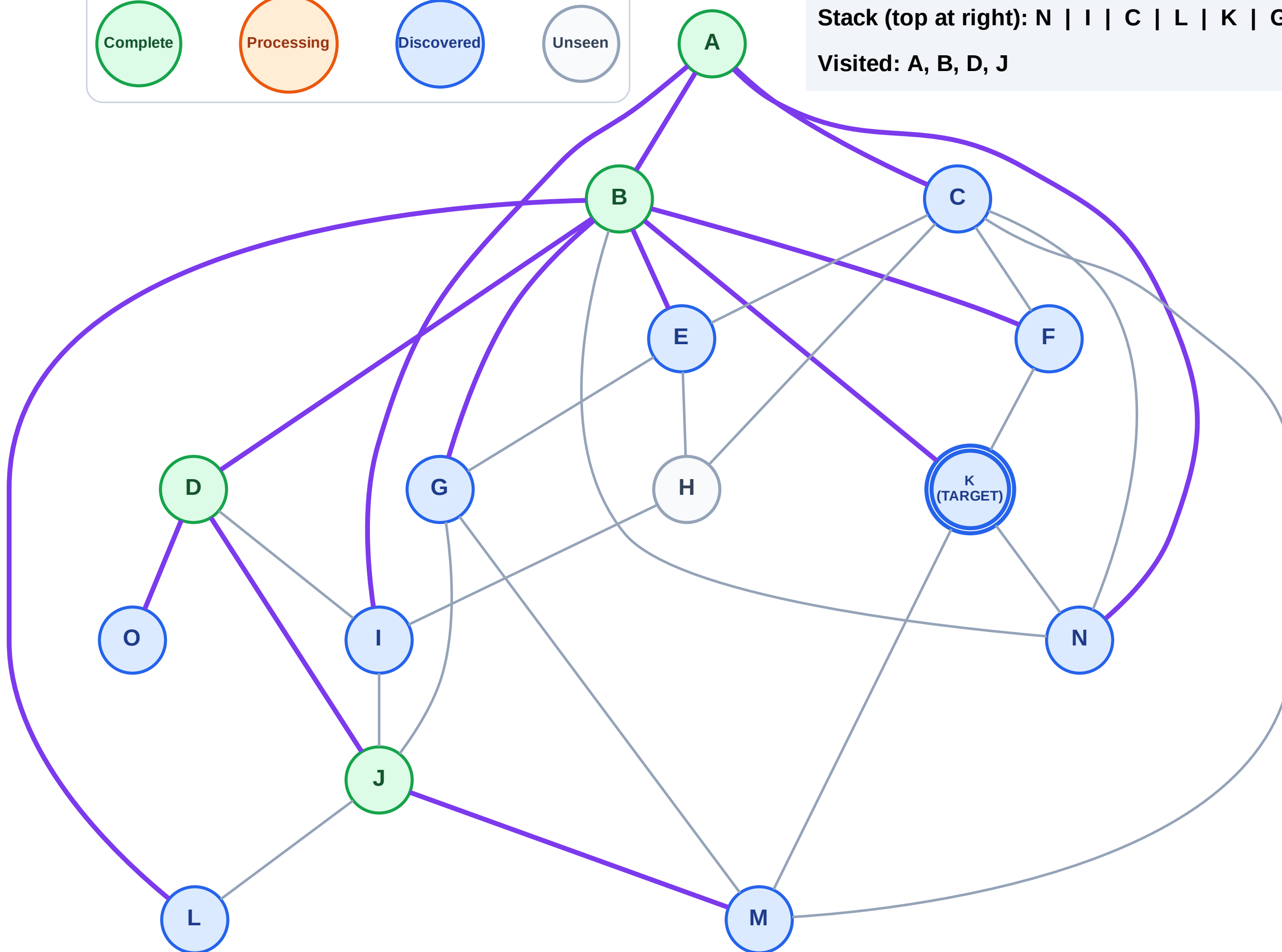
Step 9: Discover M from J

Legend



Stack (top at right): N | I | C | L | K | G | F | E | O | M

Visited: A, B, D, J



DFS Traversal

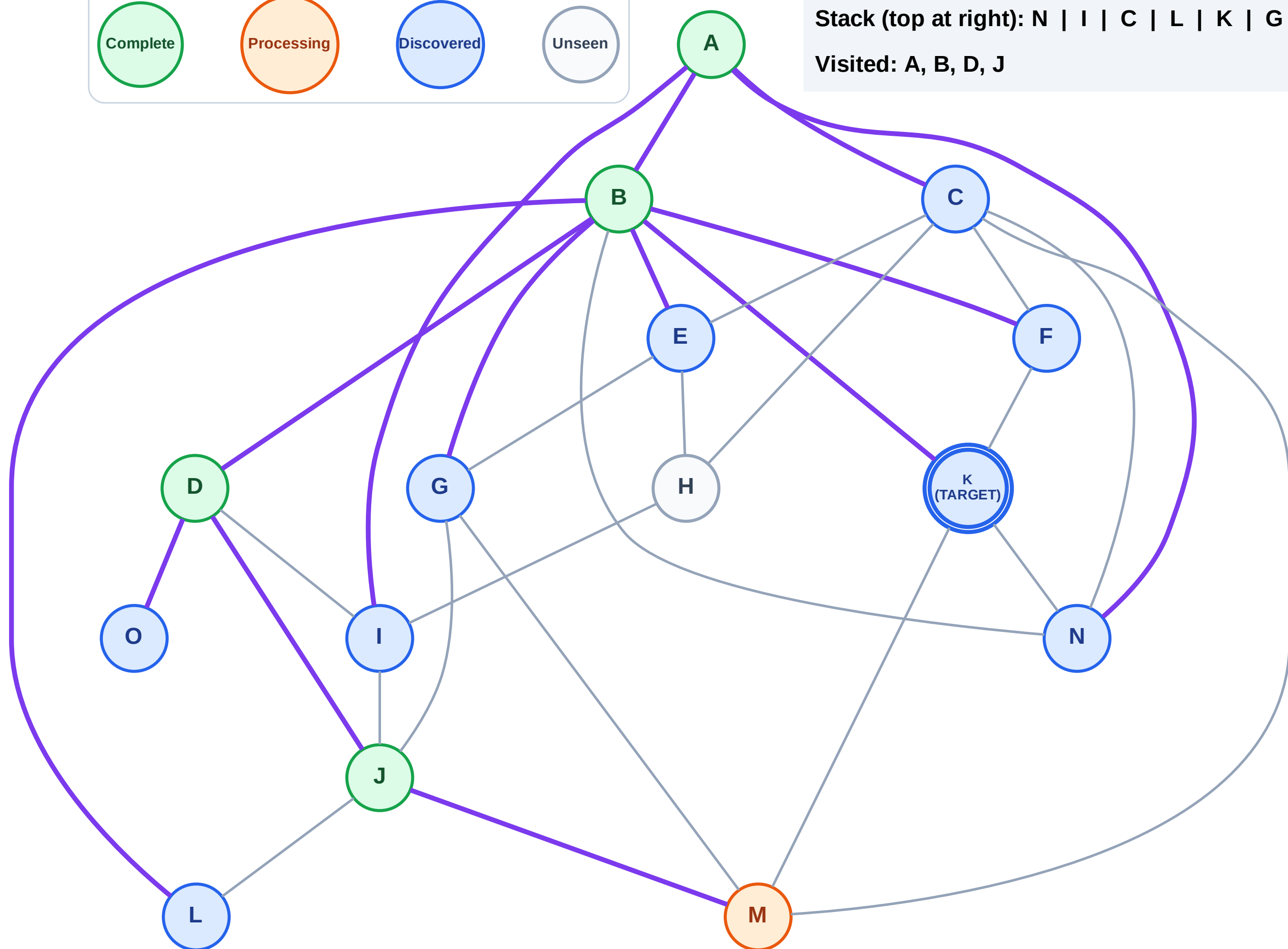
Step 10: Process node M

Legend



Stack (top at right): N | I | C | L | K | G | F | E | O

Visited: A, B, D, J



DFS Traversal

Step 11: No new neighbors from M

Legend

Complete

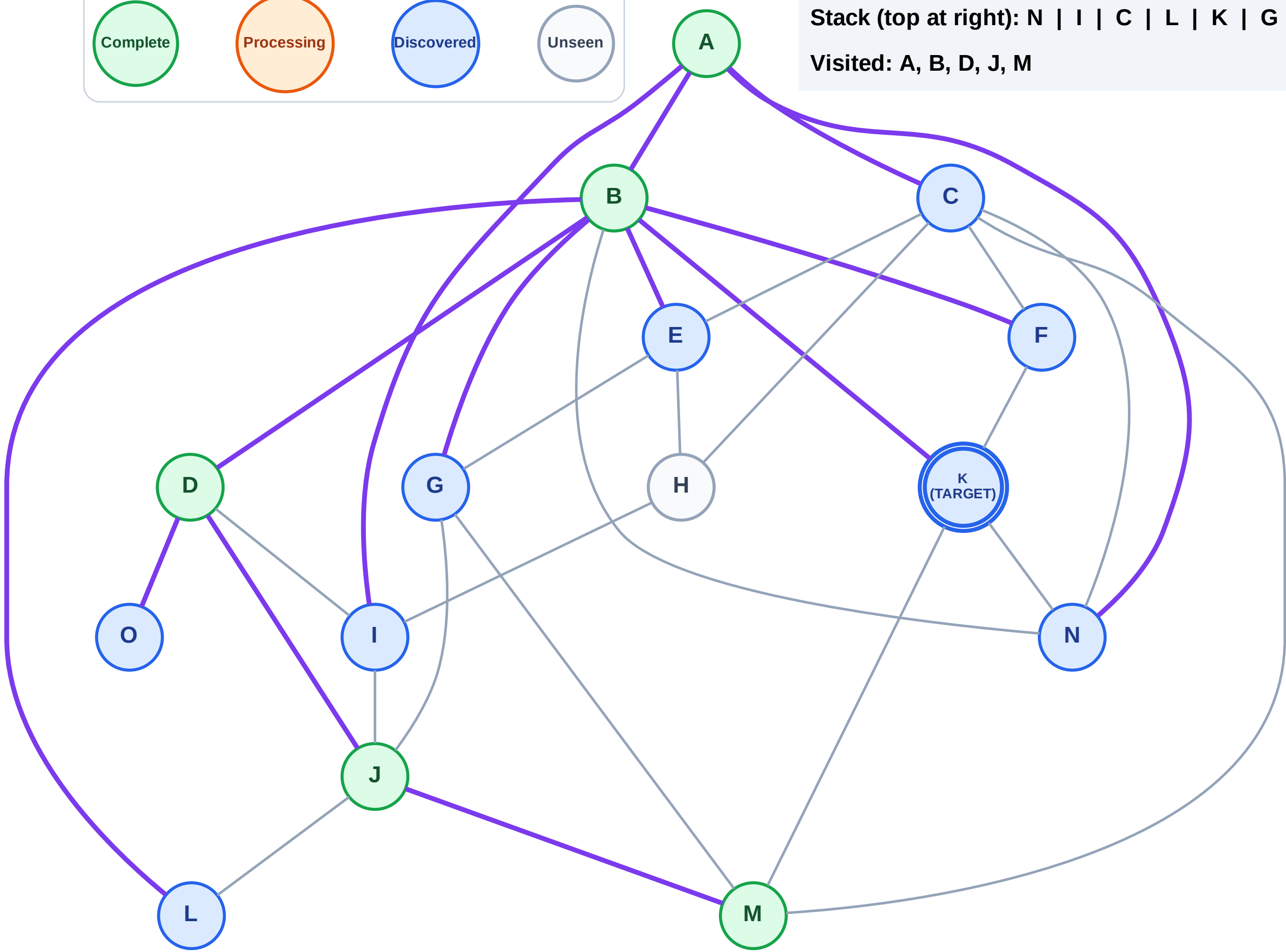
Processing

Discovered

Unseen

Stack (top at right): N | I | C | L | K | G | F | E | O

Visited: A, B, D, J, M



DFS Traversal

Step 12: Process node O

Legend

Complete

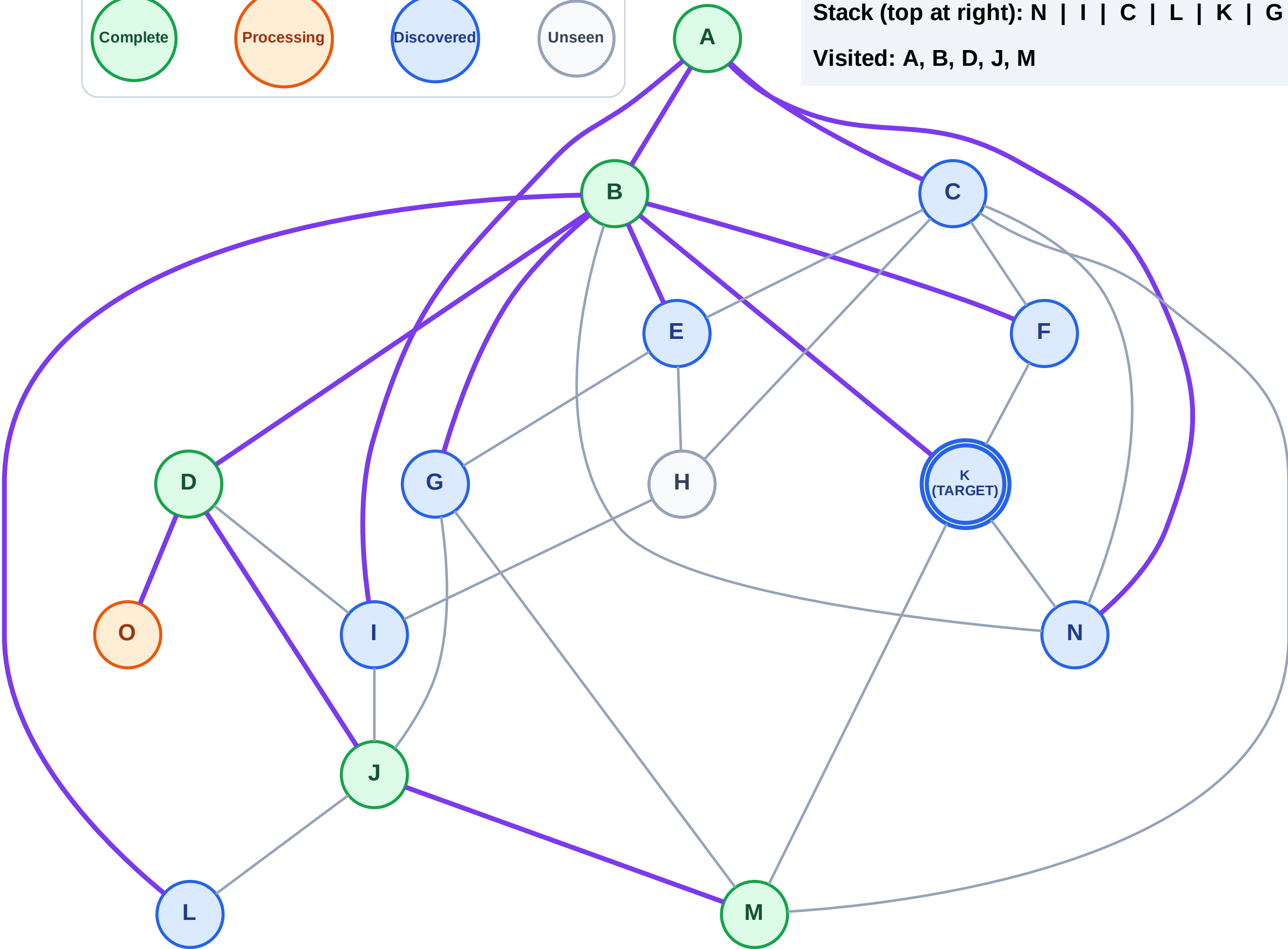
Processing

Discovered

Unseen

Stack (top at right): N | I | C | L | K | G | F | E

Visited: A, B, D, J, M

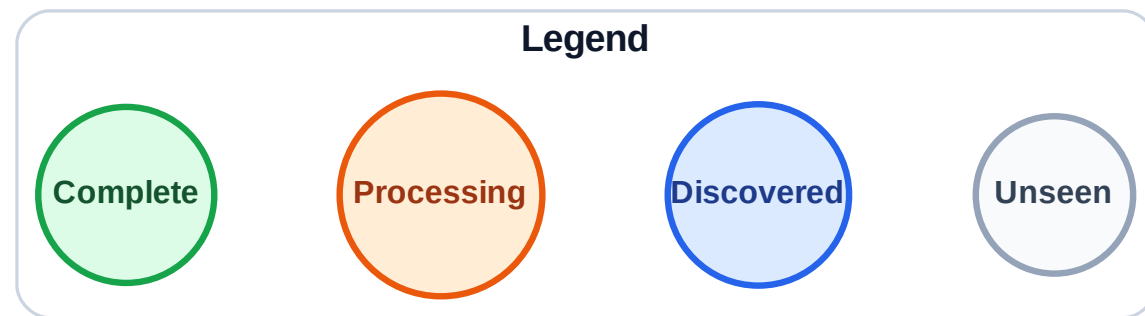


DFS Traversal

Step 13: No new neighbors from O

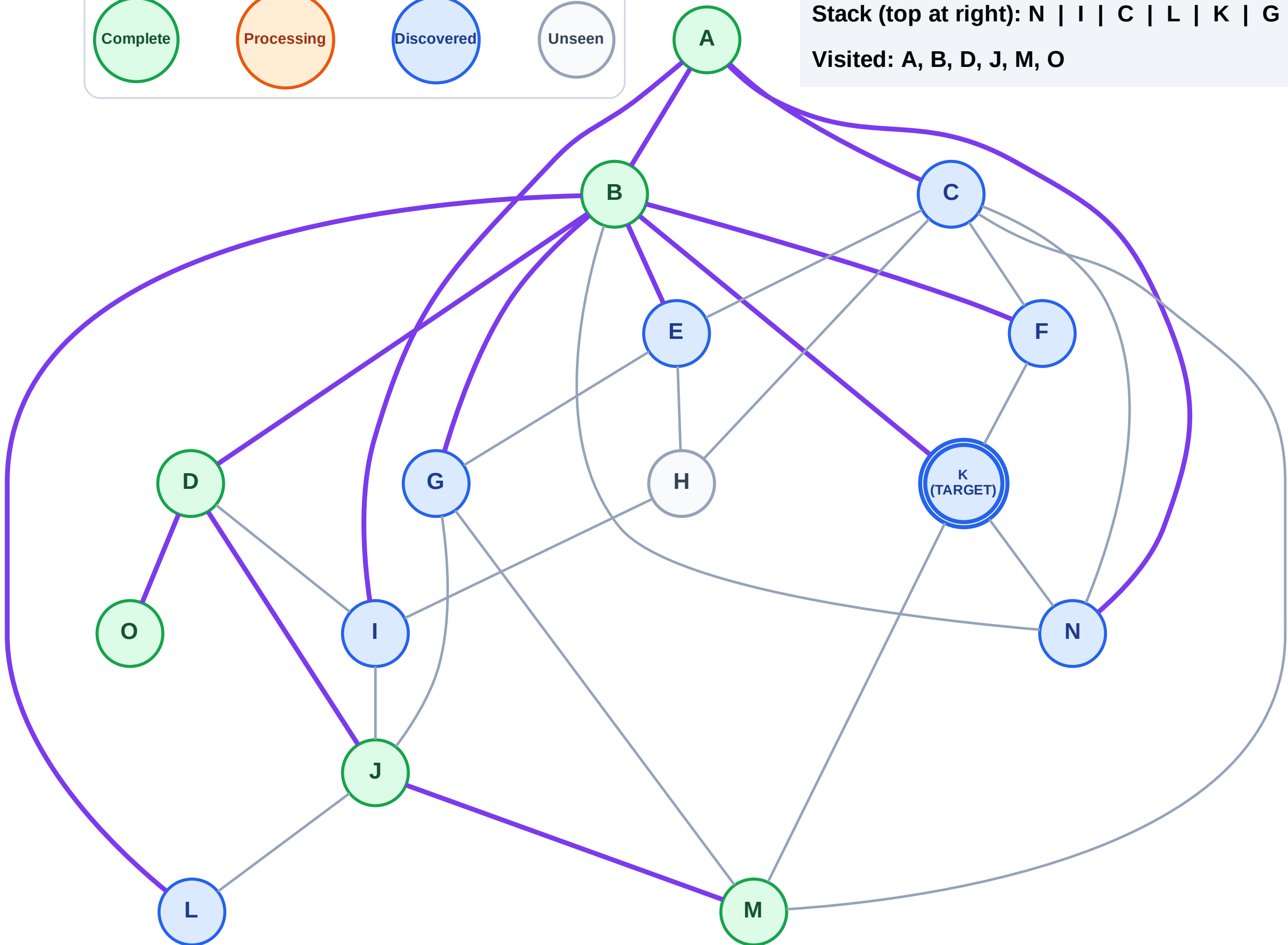
DFS Traversal

Step 13: No new neighbors from O



Stack (top at right): N | I | C | L | K | G | F | E

Visited: A, B, D, J, M, O



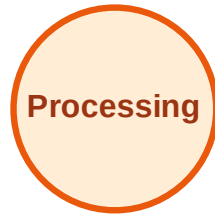
DFS Traversal

Step 14: Process node E

Legend



Complete



Processing



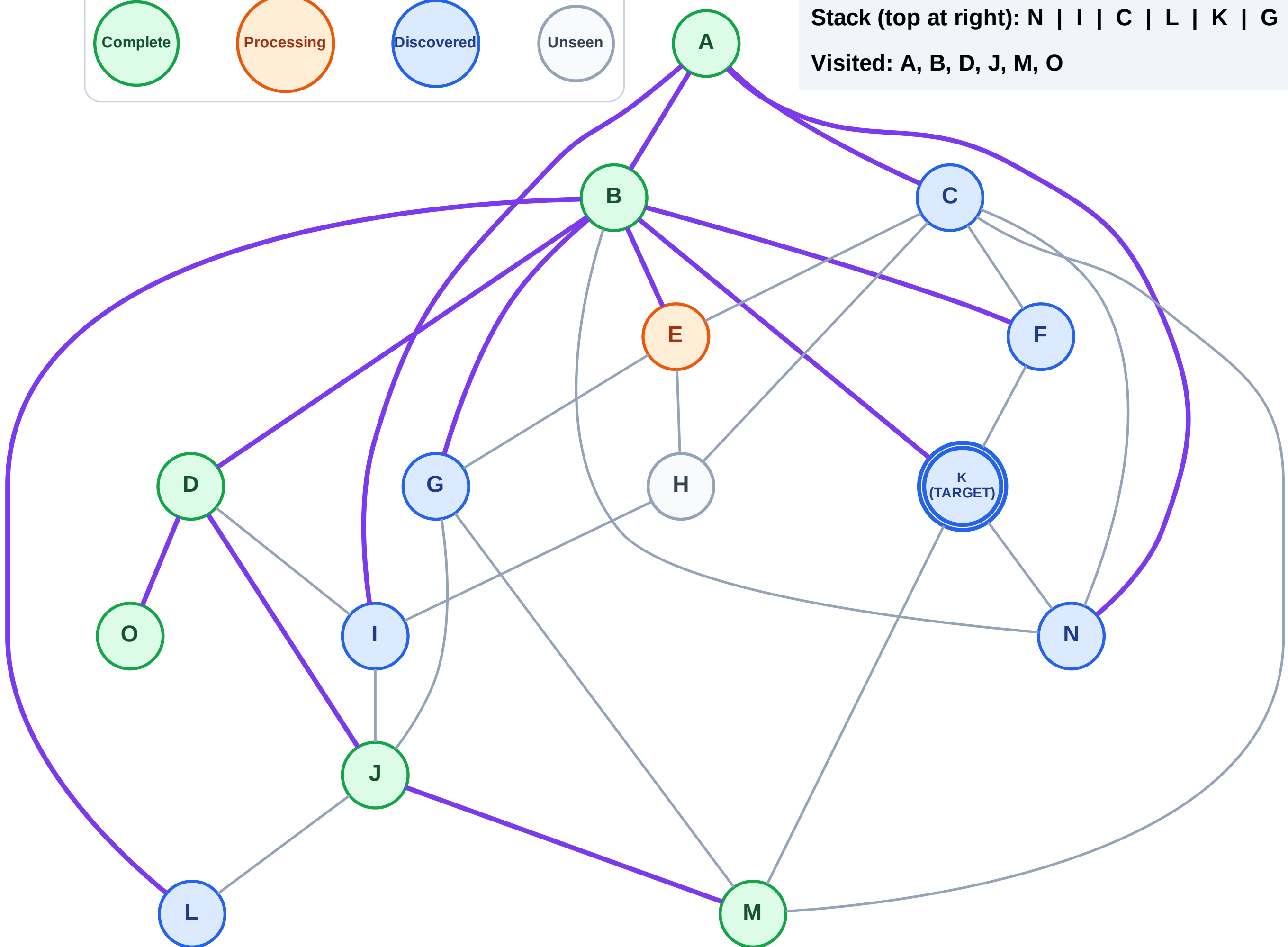
Discovered



Unseen

Stack (top at right): N | I | C | L | K | G | F

Visited: A, B, D, J, M, O



DFS Traversal

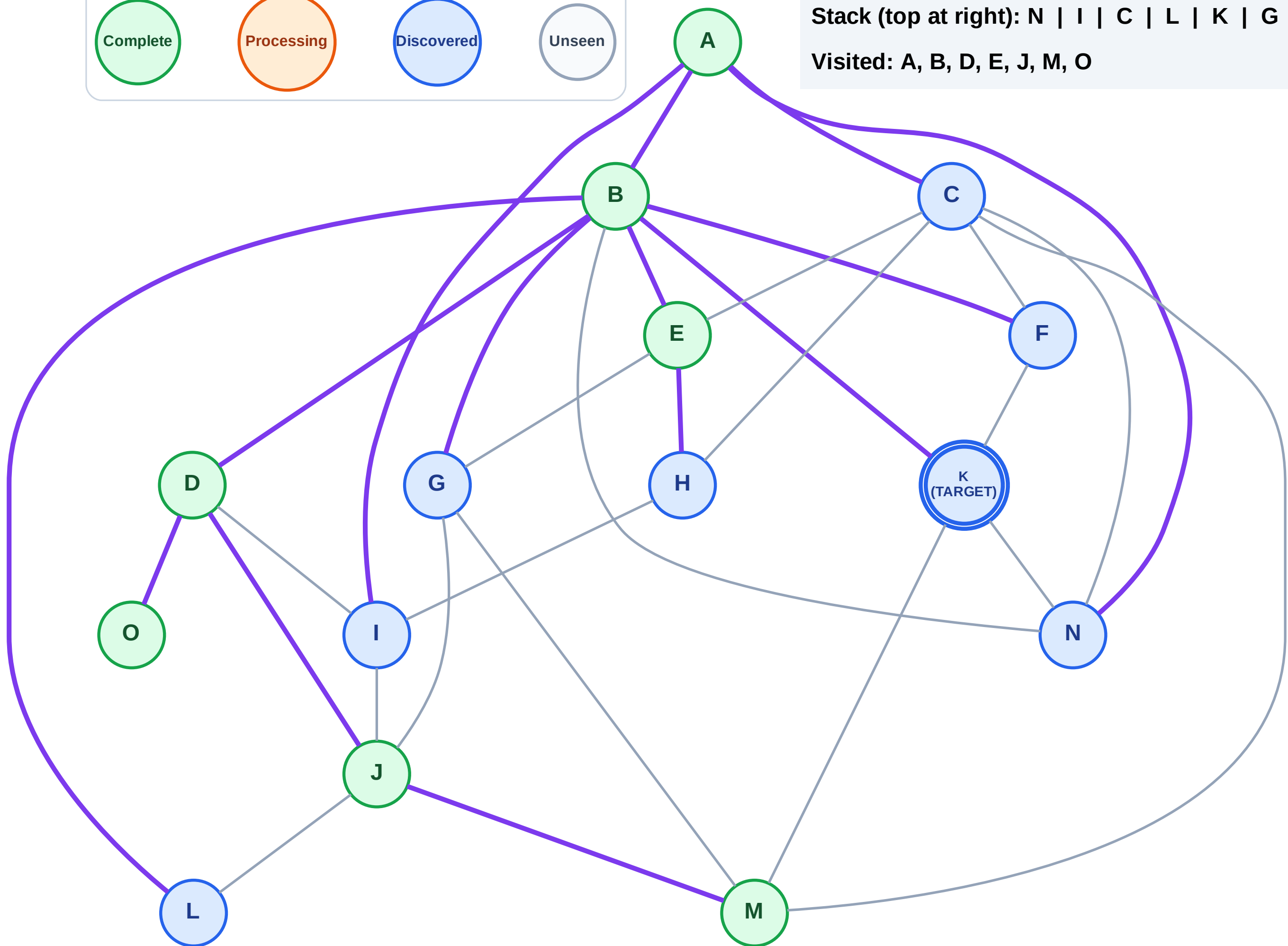
Step 15: Discover H from E

Legend



Stack (top at right): N | I | C | L | K | G | F | H

Visited: A, B, D, E, J, M, O



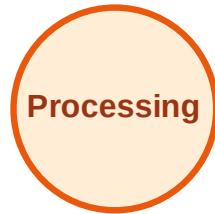
DFS Traversal

Step 16: Process node H

Legend



Complete



Processing



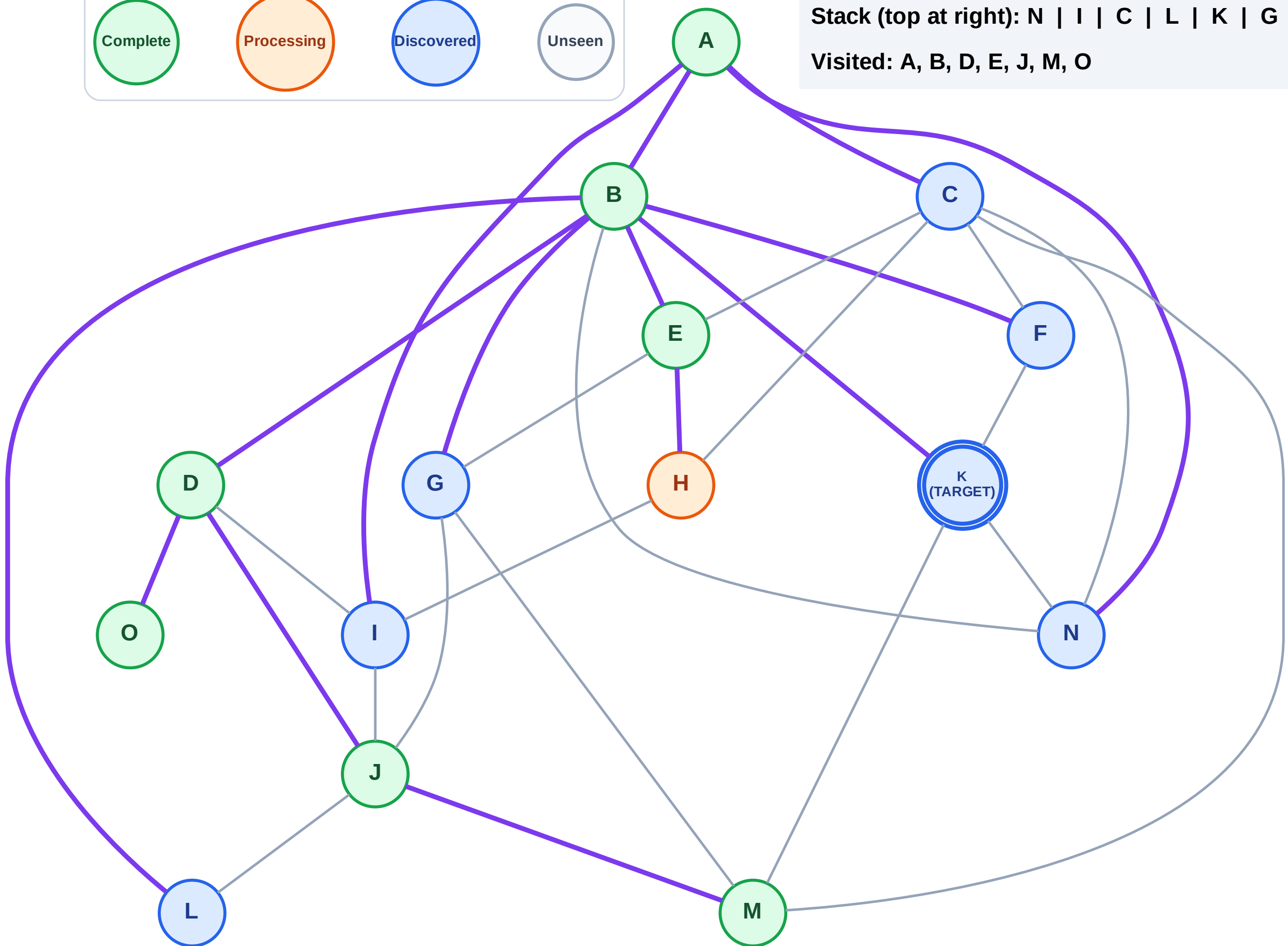
Discovered



Unseen

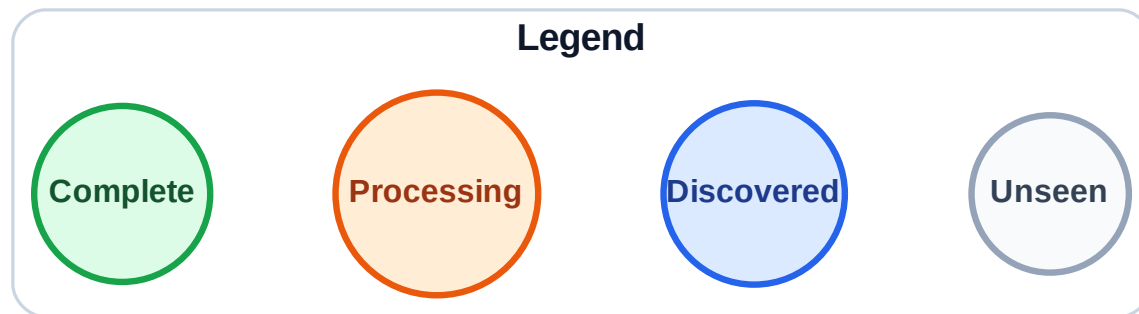
Stack (top at right): N | I | C | L | K | G | F

Visited: A, B, D, E, J, M, O



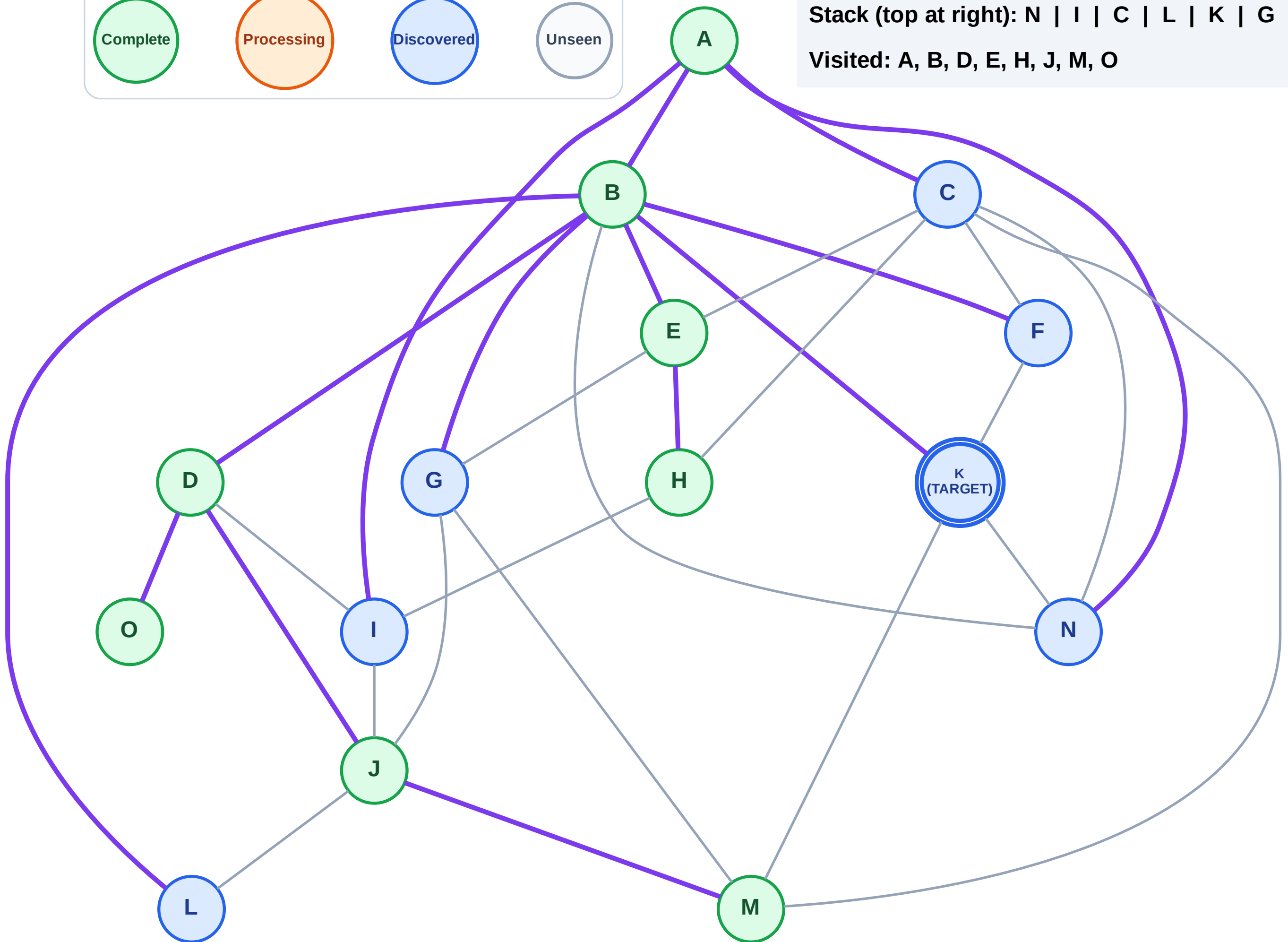
DFS Traversal

Step 17: No new neighbors from H



Stack (top at right): N | I | C | L | K | G | F

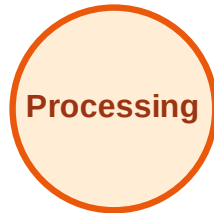
Visited: A, B, D, E, H, J, M, O



DFS Traversal

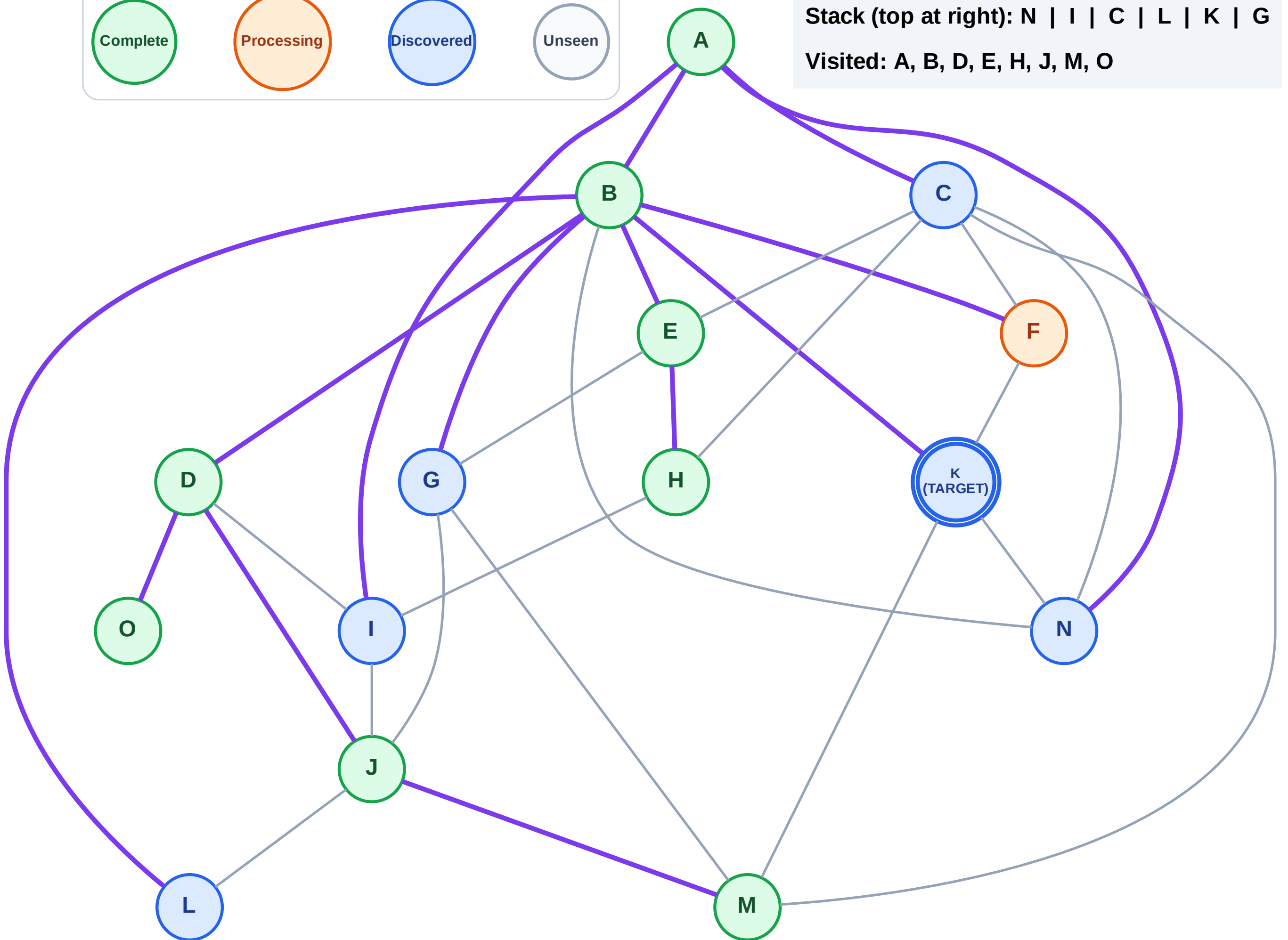
Step 18: Process node F

Legend



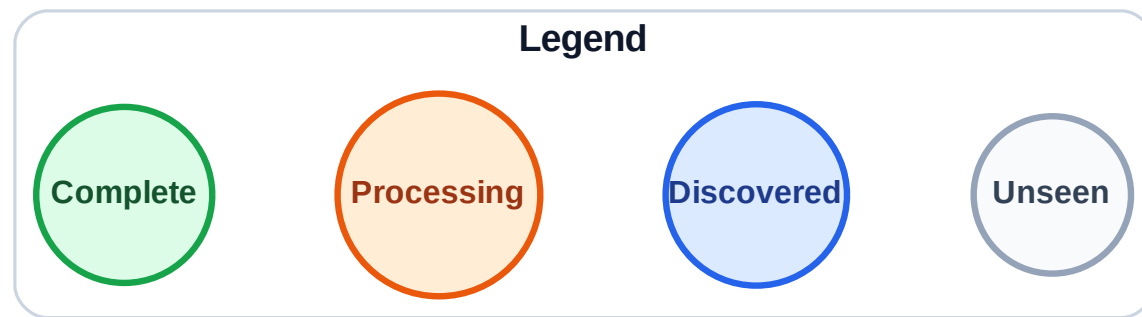
Stack (top at right): N | I | C | L | K | G

Visited: A, B, D, E, H, J, M, O



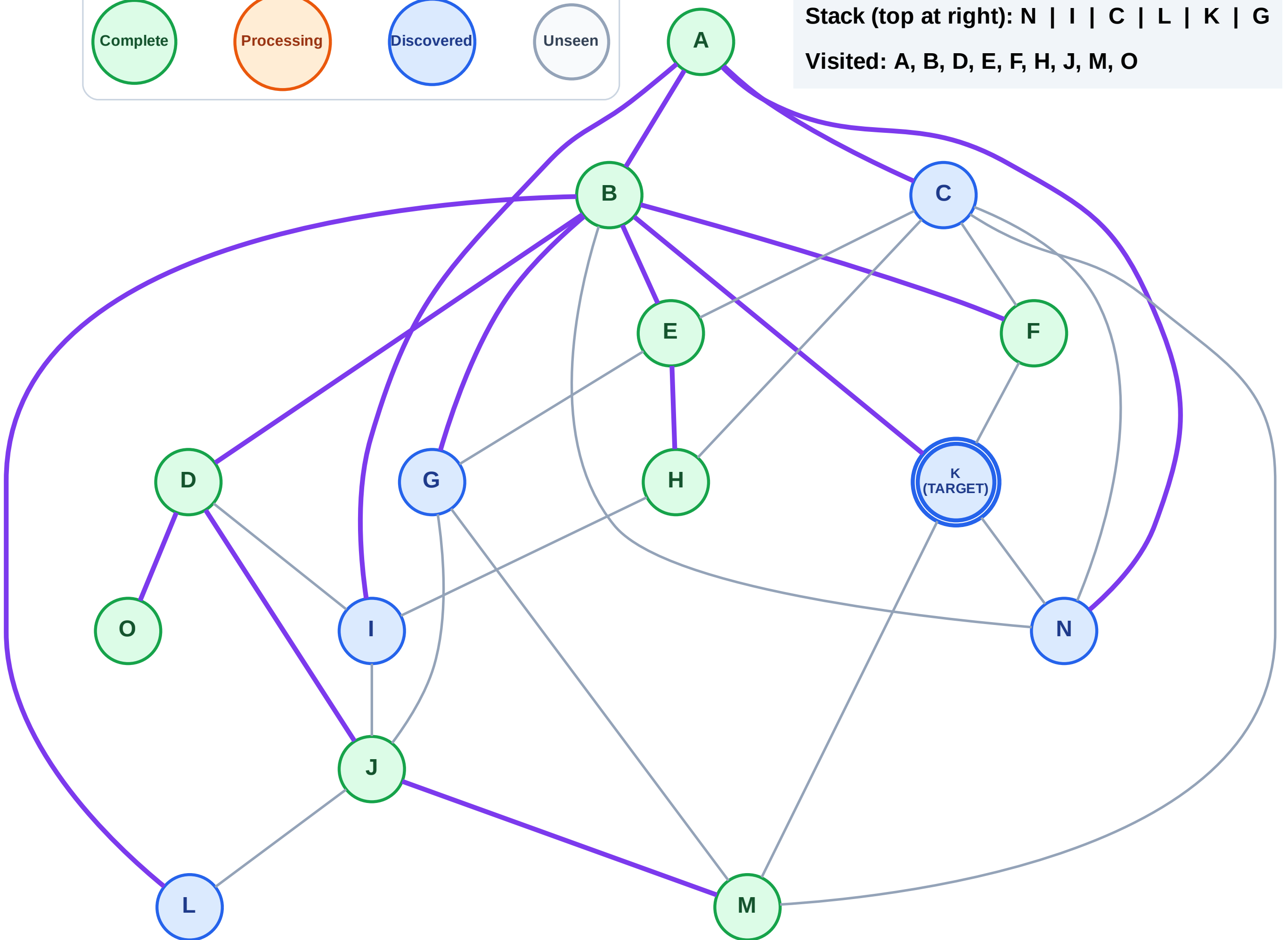
DFS Traversal

Step 19: No new neighbors from F



Stack (top at right): N | I | C | L | K | G

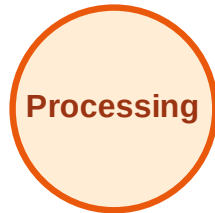
Visited: A, B, D, E, F, H, J, M, O



DFS Traversal

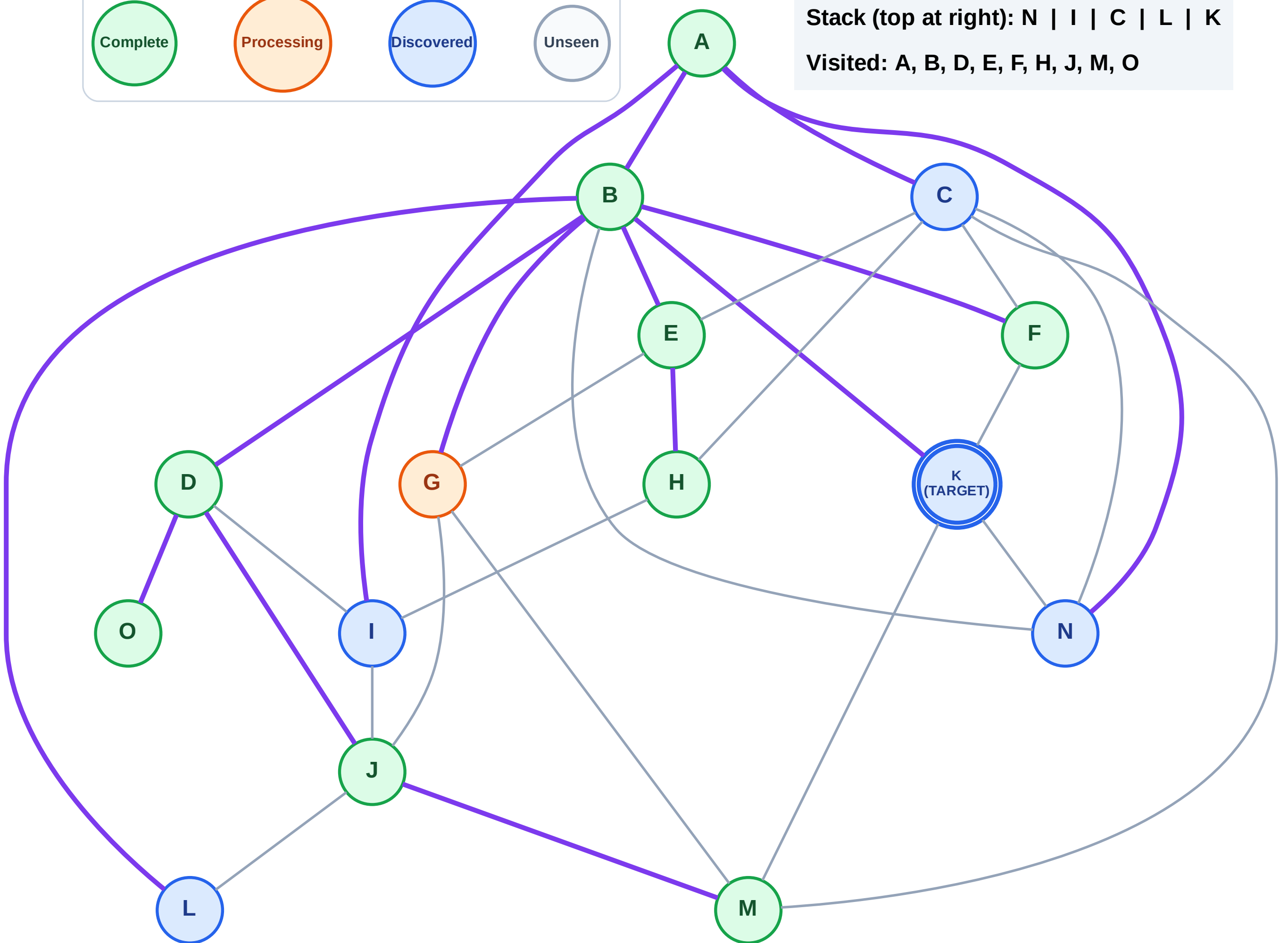
Step 20: Process node G

Legend



Stack (top at right): N | I | C | L | K

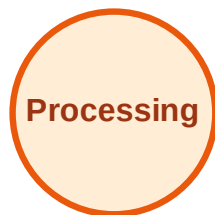
Visited: A, B, D, E, F, H, J, M, O



DFS Traversal

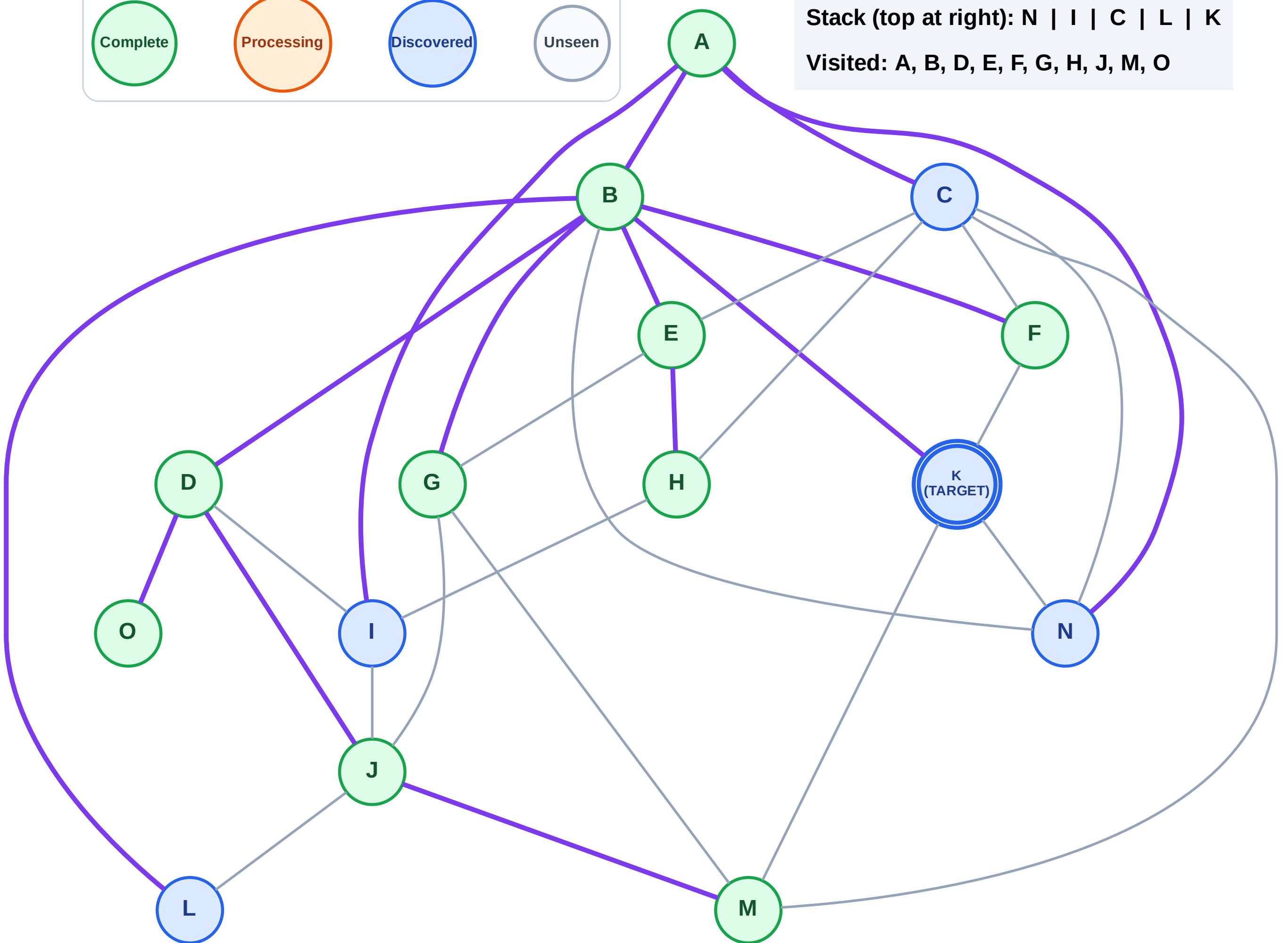
Step 21: No new neighbors from G

Legend



Stack (top at right): N | I | C | L | K

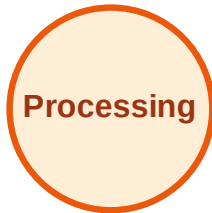
Visited: A, B, D, E, F, G, H, J, M, O



DFS Traversal

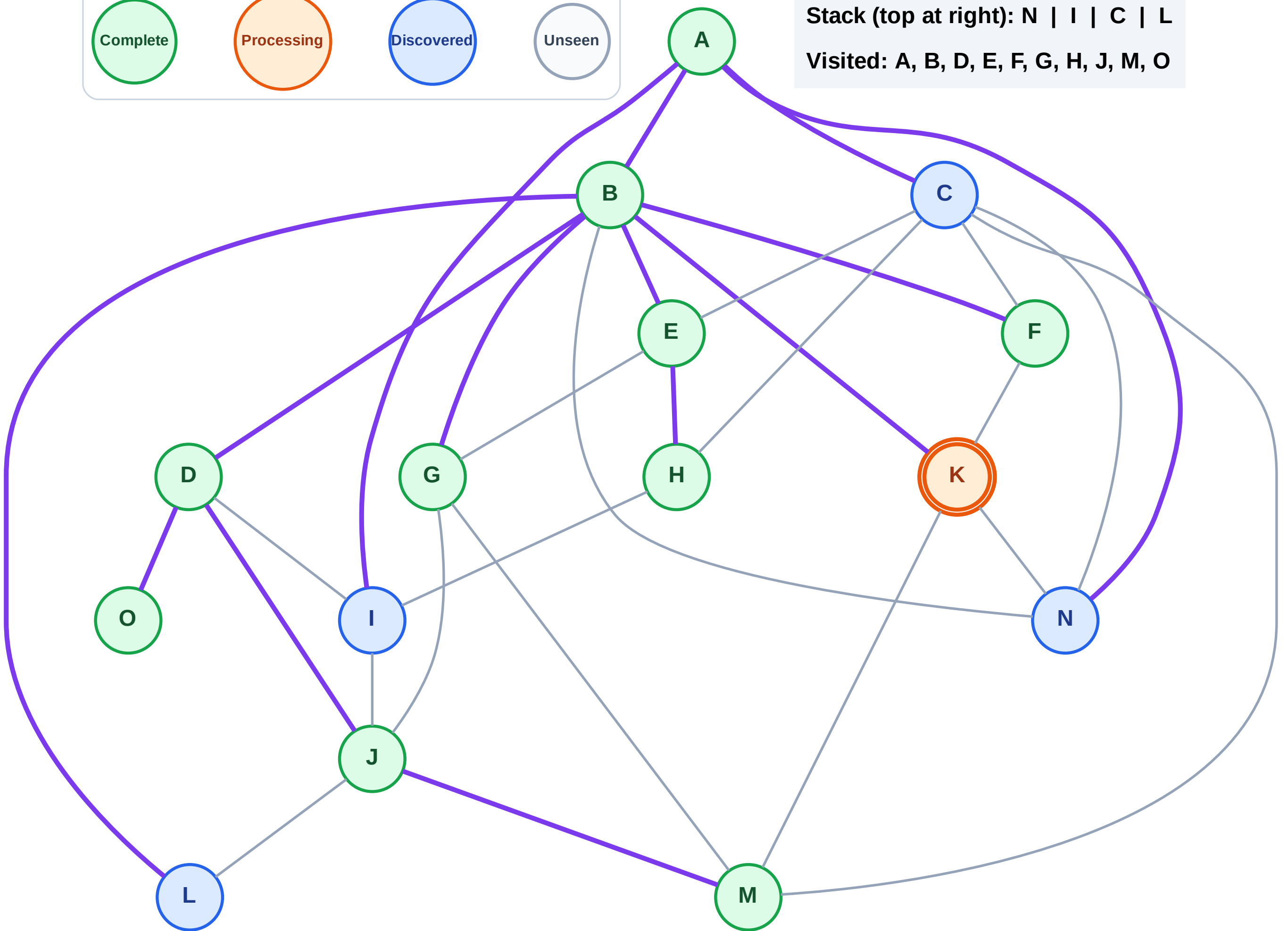
Step 22: Process node K

Legend



Stack (top at right): N | I | C | L

Visited: A, B, D, E, F, G, H, J, M, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C | L
Visited: A, B, D, E, F, G, H, J, K, M, O

